

IMAT QUESTION BANK: Biology (Medium-Hard)

130 questions · 216 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Chemistry of life and biological molecules (Q1-18)

1. The side chains of four amino acids are considered: serine bears a hydroxyl group, alanine bears a methyl group, aspartic acid bears a second carboxyl group, and lysine bears a terminal amino group. Which statement is correct?

- A) All four side chains are uncharged at pH 7.
- B) Aspartic acid is basic, because its extra carboxyl group accepts protons at pH 7.
- C) Serine and alanine belong together to the polar, uncharged class.
- D) At pH 7, aspartic acid carries a net negative charge and lysine carries a net positive charge.
- E) Lysine is acidic, because its terminal amino group releases protons readily at pH 7.

Answer: D

The decisive fact is the ionisation of each side chain at pH 7: a carboxyl side chain has already donated its proton, leaving a negative charge, while an amino side chain is still protonated and positive. Serine's hydroxyl makes it polar but uncharged, and alanine's methyl group is nonpolar, so the polar-class pairing is wrong. The tempting claim that aspartic acid is basic reverses acid behaviour, since a carboxyl group releases rather than accepts protons at neutral pH.

2. A tripeptide is assembled in the order glutamic acid, glycine, lysine. Peptide bonds are not ionisable, and the only groups carrying charge near pH 7 are the terminal amino and carboxyl groups plus the side chains of the glutamic and lysine residues. What is the net charge of this tripeptide at pH 7?

- A) -2 B) -1 C) 0 D) +1 E) +2

Answer: C

Given: tripeptide Glu-Gly-Lys at pH 7, where only the N-terminal amino group, the C-terminal carboxyl group, the glutamate side chain and the lysine side chain carry charge.

Principle: at pH 7 amino groups are protonated (+1) and carboxyl groups are deprotonated (-1), while peptide bonds carry no charge.

Step 1: N-terminal amino group = +1

Step 2: lysine side-chain amino group = +1.

Step 3: C-terminal carboxyl group = -1

Step 4: glutamate side-chain carboxyl = -1.

Step 5: net = (+1) + (+1) + (-1) + (-1) = 0

Result: net charge 0.

Classic slip: the tempting +2 counts only the two positive groups and forgets that both carboxyl groups have already given up their protons.

3. Collagen aside, proline is rarely found in the middle of an alpha helix. The best explanation is that proline:

- A)** has its side chain bonded back to the backbone nitrogen, so it cannot donate the helix hydrogen bond
- B)** cannot form peptide bonds with the residues on either side
- C)** is too small to reach across one full turn of the helix
- D)** carries a charged side chain that repels the peptide backbone
- E)** is nonpolar, and alpha helices form only around polar side chains

Answer: A

Step 1: an alpha helix is held together by hydrogen bonds in which every backbone N-H donates to a backbone C=O four residues away. Step 2: proline's side chain is bonded back to its own backbone nitrogen, so that nitrogen carries no hydrogen and cannot donate the helix hydrogen bond, and the rigid ring kinks the chain. Proline certainly can form peptide bonds, so that claim is false, and helices happily contain many nonpolar side chains such as alanine's, so polarity is not the issue.

4. A reagent breaks the four intrachain disulfide bridges of a protein made of a single polypeptide chain, without touching any peptide bond. Which levels of protein structure are directly lost, and which are preserved?

- A)** Primary and tertiary structures are lost; the secondary structure is preserved.
- B)** Quaternary structure is lost, because disulfide bridges are the bonds that hold separate chains together.
- C)** Secondary structure is directly lost; the tertiary structure is preserved.
- D)** All four levels of structure are lost, including the primary sequence.
- E)** Tertiary structure is directly lost; the primary structure is fully preserved.

Answer: E

Disulfide bridges are covalent links between side chains that lock the folded tertiary shape in place, so cutting them directly destroys the tertiary fold. The peptide bonds of the primary sequence are untouched, so the amino acid order is fully preserved, and a single polypeptide chain has no quaternary level to lose. The claim of losing the primary structure mistakes disulfide bonds for peptide bonds, and secondary structure is at most indirectly affected, not the direct target.

5. Two synthetic peptides of equal length adopt different secondary structures: peptide P forms an alpha helix and peptide Q forms an antiparallel beta sheet. Which statement is correct?

- A)** In peptide P the backbone hydrogen bonds run between neighbouring strands, while in peptide Q they run within one chain.
- B)** In peptide P the hydrogen bonds join backbone atoms within the same chain, while in peptide Q they join backbone atoms of neighbouring strands.
- C)** In peptide P the hydrogen bonds form between side chains, while in peptide Q they form along the backbone.
- D)** Neither structure uses hydrogen bonds: peptide P is held by disulfide bridges and peptide Q by ionic bonds.
- E)** Both structures are stabilised by covalent bonds between side chains.

Answer: B

Both structures are stabilised by hydrogen bonds between backbone atoms: in an alpha helix each backbone C=O donates to the N-H of the residue four positions along the same chain, while in a beta sheet the bonds run across neighbouring strands. All of these bonds are noncovalent, so options invoking covalent side-chain bonds, disulfide bridges or ionic bonds are wrong, and the version that swaps the two hydrogen-bond patterns between the structures simply reverses the facts.

6. A newly characterised protein is a long, water-insoluble strand that gives tendon its tensile strength and forms insoluble fibres outside the body. Which classification and justification are correct?

- A) Globular, because every structural protein is globular and only metabolic proteins are fibrous.
- B) Conjugated, because an insoluble protein must carry a non-protein prosthetic group.
- C) Fibrous, because all proteins found outside cells are fibrous.
- D) Fibrous, because long insoluble strands with a mechanical role are the hallmark of fibrous proteins such as collagen.
- E) Globular, because tendon proteins need a compact soluble shape to transmit force.

Answer: D

Long, water-insoluble strands with a mechanical role are the hallmark of fibrous proteins, and collagen, the tendon protein, is the textbook example. Globular proteins are the opposite: compact and soluble, like enzymes and transport proteins. The claim that all extracellular proteins are fibrous fails on albumin, a soluble globular protein working in blood plasma outside cells.

7. Which statement about levels of protein structure is correct?

- A) Quaternary structure is present in every protein that contains more than one amino acid.
- B) Replacing a single amino acid in the primary structure can change the tertiary structure of the whole chain.
- C) Hydrogen bonds between backbone atoms define the tertiary structure rather than the secondary structure.
- D) The primary structure of a protein is the ordered list of its alpha helices and beta sheets.
- E) Tertiary structure is stabilised only by the sequence and never by interactions between side chains.

Answer: B

Tertiary structure is the three-dimensional fold generated by interactions between side chains, and those side chains are dictated by the primary sequence, so a single substitution can remodel the fold of the whole chain, as in sickle-cell haemoglobin. Quaternary structure requires several separate polypeptide chains, not merely more than one residue, and hydrogen bonds between backbone atoms define secondary structure, not tertiary. The claim that tertiary structure is never stabilised by side-chain interactions is exactly backwards.

8. Cellulose and starch are both polymers of glucose, yet cellulose passes through the human gut largely undigested. The reason is that:

- A) cellulose is joined by beta-1,4 glycosidic bonds, and humans lack cellulase, the enzyme that hydrolyses them
- B) cellulose is unbranched, and no enzyme anywhere can attack an unbranched glucose chain
- C) cellulose repels water so strongly that digestive enzymes cannot reach its bonds
- D) cellulose is built from fructose units that the human gut cannot absorb
- E) cellulose molecules are too long to be broken down by any organism

Answer: A

Starch and cellulose are both glucose polymers, but starch uses alpha-1,4 glycosidic bonds while cellulose uses beta-1,4 bonds. Human amylase hydrolyses only the alpha geometry and humans secrete no cellulase, so cellulose passes through undigested. The claim that no enzyme anywhere can attack an unbranched chain is false: microbial and fungal cellulases do exactly that, which is how ruminants digest grass with the help of symbiotic microorganisms.

9. Muscle glycogen releases usable glucose units faster than the amylopectin of a starchy meal, although both are branched polymers of alpha-glucose. The best explanation is that glycogen:
- A) is first converted to fat, which diffuses through the muscle faster than glucose
 - B) contains some beta bonds that are hydrolysed faster than alpha bonds
 - C) is more highly branched than amylopectin, so many more non-reducing ends can be attacked at the same time by the mobilising enzymes
 - D) is a smaller molecule, so fewer glycosidic bonds need to be broken per gram
 - E) is partly digested by salivary amylase before it reaches the muscle

Answer: C

Glycogen phosphorylase and the debranching enzymes work only from non-reducing ends, so the speed of mobilisation tracks the number of ends available. Glycogen is far more densely branched than amylopectin, so it presents many more ends in parallel and releases glucose units faster. Both polymers are alpha-glucose polymers, so the beta-bond claim is impossible, and glycogen is mobilised inside the muscle cell itself, where salivary amylase can never reach.

10. In a chromatogram of a leaf extract, the yellow-orange bands contain carotenes and xanthophylls. Which statement about these pigments is correct?

- A) They are water-soluble pigments dissolved in the stroma of the chloroplast.
- B) They are breakdown products of chlorophyll a that have lost their magnesium atom.
- C) They are water-soluble proteins that transfer electrons to photosystem II.
- D) They are lipid-soluble pigments that absorb light only in the infrared range.
- E) They are lipid-soluble accessory pigments of the thylakoid membranes, and xanthophylls are oxygen-containing carotenoids.

Answer: E

Carotenes are pure hydrocarbons and xanthophylls are their oxygen-containing derivatives; both are lipid-soluble carotenoids embedded in the thylakoid membranes, where they act as accessory pigments broadening light capture and dissipating excess energy. They are distinct molecules, not breakdown products of chlorophyll a, and they absorb visible light, not infrared. Being lipid-soluble, they are neither water-soluble proteins nor stroma pigments.

11. Per gram, fat stores about twice as much usable energy as glycogen. Which statement explains this most directly?

- A) Fat cells store more water than glycogen deposits, and the water raises the energy yield.
- B) Glycogen is chemically a lipid, and its branching makes it less efficient.
- C) The long hydrocarbon chains of fatty acids are highly reduced, and triglycerides are stored almost without water, while glycogen binds a large mass of water.
- D) Fat is oxidised anaerobically, so none of its energy is lost as heat.
- E) Fats contain more peptide bonds per gram than carbohydrates.

Answer: C

Step 1: energy per gram rises with the degree of reduction, and the long hydrocarbon chains of fatty acids carry many more energy-rich C-H bonds per carbon than the partially oxidised glucose units of glycogen. Step 2: triglycerides are stored as almost water-free droplets, while hydrated glycogen drags a large mass of bound water, so a gram of glycogen contains less actual fuel. Fat is oxidised aerobically, so the claim of anaerobic oxidation with no heat loss is doubly wrong.

12. Starch is incubated with purified amylase under gut-like conditions, and the mixture then meets the brush-border enzymes of the small intestine. Which statement describes the carbohydrate products of the two stages correctly?

- A)** Amylase digests starch only to water and carbon dioxide; the brush-border enzymes finish the job.
- B)** Amylase mainly releases maltose and similar short chains, and the brush-border disaccharidases then release the monosaccharides.
- C)** Amylase releases free glucose from starch, and the brush-border enzymes then split glucose into smaller sugars.
- D)** Amylase converts starch into lactose, and brush-border lactase then splits the lactose.
- E)** Starch digestion cannot begin until bacteria in the large intestine supply the needed amylase.

Answer: B

Step 1: amylase hydrolyses the alpha-1,4 bonds of starch but stops at short products, so its output is maltose, maltotriose and limit dextrins rather than free glucose. Step 2: the brush-border disaccharidases (maltase, isomaltase, sucrase) then release the monosaccharides that are absorbed. The claim that amylase releases free glucose directly is the classic error, and lactose is a milk sugar that no step of starch digestion can produce.

13. Lactose, the disaccharide in milk, is hydrolysed at the intestinal brush border by lactase. The correct reaction is:

- A)** lactose + water → galactose + galactose
- B)** lactose + water → glucose + glucose
- C)** lactose + water → glucose + fructose
- D)** lactose → glucose + galactose, with water released
- E)** lactose + water → glucose + galactose

Answer: E

Lactose is a disaccharide of glucose joined to galactose, and hydrolysis splits it by adding one water molecule across the bond, so the products are exactly glucose and galactose. Glucose plus fructose is the outcome for sucrose and glucose plus glucose for maltose, and hydrolysis consumes water rather than releasing it.

14. In a tube containing enzyme and a large excess of its substrate, the enzyme concentration is doubled while temperature and pH are held constant. The initial reaction rate will:

- A)** fall to half, because the substrate is now shared among more active sites
- B)** stay the same, because enzymes are not consumed by the reaction
- C)** drop to zero, because excess substrate competes with the enzyme
- D)** double, because with substrate in excess the initial rate is proportional to the enzyme concentration
- E)** double immediately and then double again every few minutes

Answer: D

With substrate in large excess every enzyme molecule is already saturated and working at its maximum turnover, so the initial rate is proportional to the enzyme concentration: doubling the enzyme doubles the rate. Enzymes are indeed not consumed, but that only means they keep working, not that the amount present stops limiting the rate, and substrate in excess is not shared out so as to halve the rate.

15. Two tubes contain the same small, fixed amount of substrate. Tube X receives 1 unit of enzyme and tube Y receives 4 units of the same enzyme; temperature and pH are equal and optimal. Which outcome is expected?

- A)** Both tubes yield the same final amount of product, but tube Y reaches it sooner.
- B)** Tube X yields more product in the end, because its substrate lasts longer.
- C)** Neither tube produces product, because the substrate amount is below the saturation point.
- D)** Tube Y yields four times the final amount of product that tube X yields.
- E)** Tube Y reacts more slowly at first and then overtakes tube X near the end.

Answer: A

The final amount of product is fixed by the amount of substrate, which is the same in both tubes, so both tubes end with the same total product. The tube with four times as much enzyme converts the substrate faster and reaches that plateau sooner. Multiplying the final yield by four confuses rate with extent, and the given substrate amount does not prevent product formation.

16. An enzyme is added to a reaction mixture that has already reached chemical equilibrium. Which statement is correct?

- A)** The enzyme pushes the equilibrium toward the products and raises the final yield.
- B)** The enzyme lowers the activation energy of the forward direction only.
- C)** The enzyme increases the energy difference between reactants and products.
- D)** The enzyme lowers the activation energy of both directions, so equilibrium is reached sooner at the same position.
- E)** The enzyme is gradually consumed, so the mixture returns to its original state.

Answer: D

A catalyst lowers the activation energy of the forward and reverse steps equally, so both rates rise and equilibrium is reached sooner at exactly the same position. The equilibrium composition depends only on the relative energies of reactants and products, which the enzyme does not alter, so it cannot raise the final yield. Enzymes are not consumed and do not act on only one direction.

17. In prion diseases, a misfolded form of a normal membrane protein acts as a template: contact converts the normal form into the misfolded shape, which then joins growing aggregates. Which statement follows from this mechanism?

- A)** Each converted molecule becomes a new template that converts others, so the number of misfolded molecules can grow exponentially without any nucleic acid.
- B)** The misfolded protein carries a nucleic acid sequence that mutates the DNA of the cells it enters.
- C)** The aggregates are composed of lipid that has been misread from membrane phospholipids.
- D)** The reaction is self-limiting, because one misfolded molecule can convert at most one other molecule in its lifetime.
- E)** The process requires a virus to replicate, because no protein can propagate information alone.

Answer: A

Step 1: contact with a misfolded prion converts the normal membrane protein into the misfolded shape. Step 2: each newly converted molecule is itself infectious material able to convert further molecules, so the misfolded population can grow exponentially. The information transmitted is conformation alone, with no nucleic acid involved, so the claim that a virus must replicate the process is wrong, as is the model in which one misfolded molecule converts at most one other.

18. A cell line carries a mutation that disables its main chaperone proteins, while protein synthesis continues normally and amino acid sequences are unchanged. The most likely consequence inside the cell is that:

- A)** ribosomes stop translating, because chaperones supply the energy for peptide bond formation
- B)** many new chains expose their hydrophobic regions, bind each other, and aggregate instead of reaching the native fold
- C)** nothing changes, because the amino acid sequence alone always guarantees correct folding inside a living cell
- D)** the primary structure of each new protein is edited, because chaperones remove wrong amino acids during folding
- E)** new proteins are exported while still unfolded, because folding normally happens outside the cell

Answer: B

Chaperones shield exposed hydrophobic patches of new chains and give them time to reach the native fold. Without them those patches bind one another in the crowded cytoplasm, so many new proteins aggregate instead of folding. The amino acid sequence is untouched because chaperones act on conformation rather than covalent structure, and the sequence alone does not guarantee rapid correct folding against the competing reaction of aggregation.

Section B - Cells, organelles and membrane transport (Q19-34)

19. In a steroid-producing gland cell, the smooth endoplasmic reticulum is actively making hormones, while a lysosome is digesting a bacterium taken in by phagocytosis. Which pair of products is formed by the smooth endoplasmic reticulum and by lysosomal action, respectively?

- A)** Peptide hormones by the smooth endoplasmic reticulum; small soluble digestion products such as amino acids and monosaccharides by the lysosome.
- B)** Steroid hormones by the smooth endoplasmic reticulum; fully digested proteins exported out of the cell by the lysosome.
- C)** Steroid hormones by the smooth endoplasmic reticulum; small soluble digestion products such as amino acids and monosaccharides by the lysosome.
- D)** Steroid hormones by the smooth endoplasmic reticulum; hydrogen peroxide and water by the lysosome.
- E)** Steroid hormones by the rough endoplasmic reticulum; intact bacterial fragments kept permanently inside the lysosome.

Answer: C

The smooth endoplasmic reticulum is the site of lipid and steroid synthesis, so this gland cell makes steroid hormones there. Lysosomal acid hydrolases digest the phagocytosed bacterium down to small soluble molecules such as amino acids and monosaccharides, which pass into the cytosol. Peptide hormones are products of the rough endoplasmic reticulum pathway, and hydrogen peroxide plus water is the reaction of peroxisomal catalase, not of the lysosome.

20. Which of the following statements about peroxisomes and lysosomes is correct?

- A)** Peroxisomes are formed as vesicles budding from the Golgi apparatus and are filled with acid hydrolases.
- B)** Lysosomal hydrolases convert hydrogen peroxide into water and molecular oxygen.
- C)** The interior of peroxisomes is strongly acidic, which is the optimum for catalase.
- D)** Lysosomes generate hydrogen peroxide while oxidising fatty acids and amino acids.
- E)** Peroxisomal catalase decomposes the hydrogen peroxide formed inside the organelle into water and oxygen.

Answer: E

Peroxisomal oxidases produce hydrogen peroxide, and catalase inside the same organelle splits it into water and oxygen, which is the correct statement. The vesicles that bud from the Golgi and are filled with acid hydrolases are lysosomes, the strongly acidic interior belongs to the lysosome, and neither lysosomal hydrolases nor an acidic interior decompose peroxide.

21. In animal cells the first step of peroxisomal fatty acid oxidation is catalysed by an oxidase that passes electrons directly to oxygen, forming hydrogen peroxide. A mutation abolishes the activity of peroxisomal catalase. Which outcome is expected?

- A)** Fatty acid oxidation in the peroxisome stops immediately, because catalase performs the first oxidation step.
- B)** The mitochondrion takes over the decomposition of hydrogen peroxide at the cost of extra ATP consumption.
- C)** Hydrogen peroxide generated in the peroxisome is no longer decomposed, so it accumulates and can reach the cytosol, where it damages cellular components.
- D)** Hydrogen peroxide leaves the cell unchanged at a rate that fully protects the cytosol.
- E)** Hydrogen peroxide is delivered to lysosomes, where hydrolases split it safely into water and oxygen.

Answer: C

Step 1: the oxidase of the first oxidation step keeps working, so hydrogen peroxide continues to be produced. Step 2: catalase, which normally decomposes it, is absent, so peroxide accumulates. Step 3: hydrogen peroxide is small and uncharged, so it can diffuse out of the peroxisome and oxidise cytosolic components. Catalase does not perform the first oxidation step, so oxidation does not stop, and lysosomal hydrolases are not peroxide-splitting enzymes.

22. A metabolic poison halts ATP production in an animal cell kept in an isotonic medium. Which combination of transport processes continues at its normal rate for some time?

- A)** Active uptake of calcium into the endoplasmic reticulum continues, while osmosis stops.
- B)** Sodium-potassium pumping and osmosis continue, while facilitated diffusion of glucose stops.
- C)** Every carrier-mediated process stops, because all carrier proteins are driven directly by ATP hydrolysis.
- D)** Sodium-potassium pumping out of the cell continues, while osmosis through aquaporins stops.
- E)** Facilitated diffusion of glucose down its concentration gradient and osmosis of water continue, while sodium-potassium pumping stops.

Answer: E

Step 1: osmosis through aquaporins and facilitated diffusion of glucose are passive, gradient-driven processes, so both continue without ATP. Step 2: the sodium-potassium pump and the calcium pump of the endoplasmic reticulum hydrolyse ATP directly and stop. The claim that every carrier-mediated process needs ATP confuses active transport with passive carrier-mediated diffusion, and any combination keeping the sodium-potassium pump running contradicts the ATP block.

23. Cells are cultured in a medium whose glucose concentration is higher than the glucose concentration inside the cells. The cells take up glucose through a membrane carrier protein, and the uptake rate is unchanged when ATP synthesis is blocked by a poison. Which process accounts for the uptake?

- A)** Simple diffusion through the phospholipid bilayer
- B)** Facilitated diffusion through the carrier protein
- C)** Endocytosis of glucose in membrane vesicles
- D)** Secondary active transport driven by the sodium gradient
- E)** Primary active transport coupled to ATP hydrolysis

Answer: B

Glucose is polar and cannot cross the lipid bilayer by simple diffusion; it moves down its concentration gradient through a specific carrier protein and proceeds even when ATP synthesis is blocked. Carrier-mediated, down-gradient and ATP-independent is exactly facilitated diffusion. Both active-transport options would slow or stop when the poison cuts ATP, and no sodium gradient is involved here.

24. A red blood cell whose interior behaves as 300 mmol/dm³ of non-penetrating solute is placed in a large volume of solution containing 150 mmol/dm³ sodium chloride (fully dissociated) and 100 mmol/dm³ urea. Sodium chloride cannot cross the membrane; urea crosses freely. What happens to the cell?

- A)** Water first leaves the cell and it shrinks; later, water re-enters and the cell swells until it lyses.
- B)** Water enters the cell continuously from the start and the cell lyses, because the outside solution has the higher total concentration of solute particles.
- C)** The cell shrinks permanently, because urea that has entered the cytosol cannot leave again.
- D)** Water first leaves the cell and it shrinks; once urea has equilibrated across the membrane, water returns and the cell regains its original volume.
- E)** No net movement of water occurs at any time, because the sodium chloride by itself is already isotonic with the cytosol.

Answer: D

Given: cell interior = 300 mmol/dm³ of non-penetrating solute; outside = 150 mmol/dm³ NaCl, fully dissociated and unable to cross, plus 100 mmol/dm³ urea, which crosses freely; outside volume so large that its concentrations barely change.

Principle: water follows the total particle concentration, but only non-penetrating solutes exert a lasting osmotic pull.

Step 1: outside particles at the start = $(150 \times 2) + 100 = 400$ against 300 inside, so water leaves and the cell shrinks.

Step 2: urea diffuses in down its gradient until both sides hold 100.

Step 3: at equilibrium each side holds 300 fixed particles + 100 urea = 400 particles in total.

Step 4: the lasting pull now comes only from the non-penetrating solutes: 300 outside vs 300 inside, so net water movement stops with the cell back at its original volume.

Result: transient shrinkage, then complete recovery of the original volume.

Classic slip: predicting lysis ignores that urea, once equilibrated, cancels its own osmotic effect on both sides equally.

25. Which statement about the fluid mosaic model of the plasma membrane is correct?

- A)** Phospholipids and many membrane proteins can move laterally within the bilayer, which behaves as a two-dimensional fluid.
- B)** Cholesterol is the main component of plant-cell membranes and is absent from animal cells.
- C)** The membrane is a rigid structure in which every protein is fixed in one position for the life of the cell.
- D)** The core of the bilayer is hydrophilic, which is why small ions cross it freely.
- E)** The hydrophobic fatty acid tails of the phospholipids face the aqueous solutions on both sides of the membrane.

Answer: A

In the fluid mosaic model the bilayer is a two-dimensional fluid in which phospholipids and many membrane proteins diffuse laterally, with the proteins forming a mosaic. The hydrophobic tails face each other in the interior, not the water, the hydrophobic core blocks small ions rather than letting them cross freely, and cholesterol is a characteristic component of animal-cell membranes, not of plant membranes and not absent from animals.

26. Researchers treat sealed vesicles of plasma membrane in two ways. Treatment 1 uses a concentrated salt solution that disrupts ionic and polar interactions but leaves the bilayer intact. Treatment 2 uses a mild detergent that dissolves the bilayer. Which result is expected?

- A)** Treatment 1 releases peripheral proteins bound to the membrane surfaces, while treatment 2 releases integral proteins such as transmembrane channels.
- B)** Treatment 1 releases both classes of protein equally, because ionic interactions anchor the integral proteins in the bilayer.
- C)** Neither treatment can release integral proteins, because transmembrane segments cannot be separated from lipids.
- D)** Both treatments release only peripheral proteins, because integral proteins are protected by carbohydrate chains.
- E)** Treatment 1 releases integral proteins, because their transmembrane segments bind ions, while treatment 2 releases only peripheral proteins.

Answer: A

Peripheral proteins cling to the membrane surfaces through ionic and polar interactions, so a concentrated salt solution strips them off while the bilayer stays intact. Integral proteins are anchored by hydrophobic transmembrane segments buried in the bilayer, so only a detergent that dissolves the bilayer frees them. The claim that ionic interactions anchor the integral proteins reverses the chemistry: their anchor is hydrophobic, not ionic.

27. A red blood cell is moved from an isotonic medium into a solution with a higher total concentration of non-penetrating solutes than the cytosol. What is the outcome?

- A)** Water enters the cell by osmosis and the cell swells.
- B)** Water leaves the cell by osmosis and the cell shrivels.
- C)** Water leaves the cell until the membrane bursts.
- D)** The volume stays constant, because a cell wall resists any change in water content.
- E)** Non-penetrating solutes enter the cell until the two concentrations are equal, so the volume stays constant.

Answer: B

A medium with a higher concentration of non-penetrating solutes is hypertonic, so water leaves the red blood cell by osmosis and it shrivels. Swelling and bursting belong to the opposite, hypotonic case, red blood cells have no cell wall, and non-penetrating solutes by definition cannot enter to equalise the concentrations.

28. Cells from the same tissue are divided into two samples: one sample is placed in solution X, which is hypertonic to the cells, and the other in solution Y, which is hypotonic to the cells. Both plant and animal cells are treated this way. Which statement about the outcomes is correct?

- A)** In solution X the animal cells remain unchanged because their membrane is elastic, while in solution Y the plant cells burst.
- B)** In solution X the animal cells become turgid, while in solution Y the plant cells show plasmolysis.
- C)** In solution X the plant cell walls collapse inward together with the membrane, while in solution Y the animal cells become turgid.
- D)** In solution X the plant cells show plasmolysis, with the plasma membrane pulling away from the intact wall, while in solution Y the animal cells swell and may burst.
- E)** In solution X both cell types burst, while in solution Y both cell types shrink.

Answer: D

In the hypertonic solution water leaves both cell types: the plant protoplast shrinks and the membrane pulls away from the rigid wall (plasmolysis), which itself does not collapse, while the animal cell simply shrivels. In the hypotonic solution water enters: the wall lets the plant cell build turgor pressure without bursting, whereas the wall-less animal cell swells and may lyse. Turgor belongs to plant cells in hypotonic media, so any description assigning turgor or collapse to the wrong cell type or solution is wrong.

29. Viruses are described as acellular. Which statement gives the main reason?

- A)** A virus particle is far too small to be seen with a light microscope.
- B)** A virus can be crystallised and stored like a chemical substance.
- C)** A virus has no cell wall around its nucleic acid.
- D)** A virus particle contains either DNA or RNA, but never both at once.
- E)** A virus has no cytoplasm, no ribosomes and no metabolism of its own, and it can multiply only inside a host cell.

Answer: E

A virus has no cytoplasm, no ribosomes and no metabolism of its own and can multiply only inside a host cell, so it lacks the defining features of a cell and is acellular. Containing DNA or RNA but never both is a true property but not the reason it is not cellular, and small size or the ability to be crystallised say nothing about cellularity.

30. Which of the following statements about viruses is correct?

- A)** Viruses increase in number by binary fission within the host cytoplasm.
- B)** The envelope of an enveloped virus is assembled on its own ribosomes from lipids synthesised by viral enzymes.
- C)** An enveloped virus takes its lipid envelope from a host-cell membrane as it leaves the cell and inserts its own glycoproteins into that envelope.
- D)** Viruses carry ribosomes that switch on only after the nucleic acid has entered the host cell.
- E)** A single virus particle may contain one DNA genome and one RNA genome at the same time.

Answer: C

An enveloped virus takes its lipid envelope from a host-cell membrane as it buds out or leaves by exocytosis, and inserts virus-encoded glycoproteins into that lipid. Viruses never carry DNA and RNA genomes at the same time, possess no ribosomes at all, and increase in number by assembly of new particles rather than by binary fission, which is a cellular process.

- 31.** Which pairing of a plastid type with what it stores is correct?
- A)** Aleuroplasts store oils and lipids, for example in oil seeds.
 - B)** Chromoplasts store chlorophyll, for example in mature leaves.
 - C)** Elaioplasts store proteins, for example in cereal grains.
 - D)** Amyloplasts store the red and yellow pigments, for example in tomato fruits.
 - E)** Amyloplasts store starch, for example in potato tubers.

Answer: E

Amyloplasts are the starch-storing leucoplasts, the classic example being potato tubers. Elaioplasts store oils and lipids, aleuroplasts store protein, chromoplasts hold the red and yellow carotenoid pigments of fruits and petals, and chlorophyll sits in chloroplasts, so the pairings that swap oil and protein storage, or place pigments in amyloplasts or chlorophyll in chromoplasts, are all wrong.

- 32.** During the ripening of a tomato fruit the green colour fades, red pigments accumulate, and the starch stored in the fruit is converted into sugars. Which combination of plastid events fits these changes?
- A)** Chromoplasts differentiate back into chloroplasts as the fruit ripens, while elaioplasts hydrolyse the starch.
 - B)** Chloroplasts differentiate into leucoplasts that store the red pigments, while amyloplasts build new starch from the sugars.
 - C)** Chloroplasts differentiate into amyloplasts as starch accumulates, while chromoplasts break the sugars down further.
 - D)** Chloroplasts differentiate into chromoplasts as carotenoids accumulate, while amyloplasts release the starch that is hydrolysed to sugars.
 - E)** Elaioplasts differentiate into chloroplasts as chlorophyll returns, while aleuroplasts release carotenoids.

Answer: D

Step 1: as the fruit ripens, chloroplasts differentiate into chromoplasts, which accumulate the red carotenoid lycopene. Step 2: the starch held in amyloplasts is hydrolysed to sugars, sweetening the tissue. The reversed conversions get the direction of differentiation wrong, red pigments are stored in chromoplasts rather than in leucoplasts, and new starch is not being built during ripening.

- 33.** In Tay-Sachs disease, a defective gene leaves one enzyme inactive inside a particular organelle. Nerve cells of affected children develop swollen vesicles packed with undigested ganglioside lipids. Which statement identifies the organelle and explains the accumulation?
- A)** The lysosome lacks a functional hydrolytic enzyme, so lipid delivered to it for digestion is not broken down and accumulates inside it.
 - B)** The peroxisome lacks catalase, so lipids taken in for oxidation accumulate inside it.
 - C)** The Golgi apparatus fails to package secretory proteins, so undigested lipid collects in secretory vesicles.
 - D)** The mitochondrion cannot oxidise the gangliosides, so the lipid is stored inside mitochondrial cristae.
 - E)** The smooth endoplasmic reticulum overproduces lipids faster than they can be exported, so lipid-filled vesicles swell up.

Answer: A

Tay-Sachs is a lysosomal storage disease: a defective acid hydrolase inside the lysosome cannot degrade ganglioside lipids, so the lipid delivered there for digestion accumulates and swells the vesicles. Peroxisomes oxidise fatty acids but do not digest gangliosides, and the endoplasmic reticulum and Golgi carry out synthesis and packaging steps, not terminal digestion, so none of those options identifies the lesion.

34. Three treatments each disable one organelle in a cell line. After treatment 1, secreted proteins are synthesised but are never given mature carbohydrate chains and are not sorted to their destinations. After treatment 2, the cells cannot break down very-long-chain fatty acids. After treatment 3, engulfed bacteria are not digested even though acid hydrolases are produced normally. Which set of organelles is disabled by treatments 1, 2 and 3, respectively?

- A)** Rough endoplasmic reticulum; peroxisome; lysosome
- B)** Golgi apparatus; peroxisome; lysosome
- C)** Rough endoplasmic reticulum; lysosome; peroxisome
- D)** Smooth endoplasmic reticulum; peroxisome; Golgi apparatus
- E)** Golgi apparatus; lysosome; peroxisome

Answer: B

Treatment 1: maturation of carbohydrate chains and sorting of proteins to their destinations are Golgi functions. Treatment 2: very-long-chain fatty acids are broken down in peroxisomes. Treatment 3: engulfed bacteria are digested by lysosomal acid hydrolases, which are still produced, so the lesion is in the lysosome itself, giving Golgi, peroxisome, lysosome in that order. The option assigning the rough endoplasmic reticulum to treatment 1 overweighs the initial glycosylation step that happens there, but mature chains and sorting are Golgi work.

Section C - Molecular genetics and biotechnology (Q35-48)

35. A bacterial culture is treated with a drug that completely inhibits DNA gyrase, a topoisomerase. Helicase and DNA polymerase are unaffected, and replication was already under way at several forks when the drug was added. Which outcome is expected?

- A)** Helicase can no longer separate the two strands, because topoisomerase is the enzyme that breaks the hydrogen bonds between complementary bases.
- B)** Okazaki fragments accumulate on the lagging strand, because topoisomerase normally seals the nicks between adjacent fragments.
- C)** Replication forks stall shortly after drug addition, because unwinding by helicase leaves the unreplicated DNA ahead of each fork increasingly overwound and torsionally strained.
- D)** Replication continues at normal speed, because topoisomerase acts only at the very end of replication to link the two daughter molecules.
- E)** The separated parental strands cannot act as templates, because topoisomerase proofreads newly synthesized DNA.

Answer: C

Step 1: as helicase unwinds the parental duplex, the unreplicated DNA ahead of each fork becomes increasingly overwound and torsionally strained, since the ends of a circular bacterial chromosome cannot rotate freely. Step 2: DNA gyrase, a topoisomerase, exists precisely to relieve that strain. Step 3: with gyrase fully inhibited the strain builds until the fork machinery can no longer advance, so replication forks stall shortly after drug addition even though helicase and polymerase are intact. Gyrase neither breaks the hydrogen bonds between bases (helicase does) nor seals nicks between Okazaki fragments (ligase does).

36. In a temperature-sensitive bacterial mutant shifted to the restrictive temperature, DNA ligase is inactive while helicase, primase and DNA polymerase still function normally. Which observation is expected in cells that were replicating at the time of the shift?

- A)** RNA primers are no longer removed from the fragments, because primer removal is carried out by DNA ligase.
- B)** The lagging strand remains as a series of separate Okazaki fragments, because the nicks between adjacent fragments are not sealed.
- C)** The leading strand also remains in fragments, because ligase normally cuts the leading strand into pieces during replication.
- D)** No new Okazaki fragments are synthesized at all, because the synthesis of each fragment requires ligase.
- E)** Helicase stops unwinding the parental double helix, because the torsional stress generated by unwinding can only be resolved by ligase.

Answer: B

Primase and DNA polymerase are intact, so Okazaki fragments are still synthesised on the lagging strand; ligase's only role is forming the final phosphodiester bond that seals the nick between adjacent fragments. With ligase inactive the lagging strand therefore stays as a series of separate fragments. Torsional stress is resolved by topoisomerase, primer removal is not ligase's task, and the leading strand is made continuously.

37. In a eukaryotic species, nucleosomes are arranged with a repeat length of 200 bp of DNA per nucleosome. Each nucleosome core particle contains two copies each of histones H2A, H2B, H3 and H4, with 147 bp of DNA wrapped around the octamer. A linear stretch of this chromatin is 24,000 bp long. How many molecules of histone H3 are present in its nucleosome core particles?

- A)** 60 molecules **B)** 120 molecules **C)** 240 molecules **D)** 480 molecules **E)** 960 molecules

Answer: C

Given: repeat length 200 bp per nucleosome; each core particle holds two copies each of H2A, H2B, H3 and H4; stretch length 24,000 bp.

Principle: count core particles with the repeat length, because that is the spacing of nucleosomes along the DNA (the 147 bp wrap is detail inside each particle).

Step 1: core particles = $24,000 \div 200 = 120$

Step 2: H3 copies per core particle = 2

Step 3: total H3 molecules = $2 \times 120 = 240$

Result: 240 molecules of histone H3.

Classic slip: 480 credits all four core histone types with an H3 copy, and 960 treats every protein of the octamer as H3.

38. Which statement about histones is correct?

- A)** Histones are small basic proteins whose positive charges bind the negatively charged phosphate groups of the DNA backbone.
- B)** Each nucleosome core particle contains eight different histone types, one molecule of each kind.
- C)** Histones are covalently joined to DNA by a ligation reaction, which is what holds each nucleosome together.
- D)** Bacterial chromosomes are organized around histone octamers in the same way as eukaryotic chromosomes.
- E)** Histones are acidic proteins that bind DNA through attraction between their negatively charged side chains and the positively charged DNA backbone.

Answer: A

Histones are small proteins rich in lysine and arginine, so they are basic; their positive charges bind ionically to the negative phosphate groups of the DNA backbone. No covalent ligation joins histones to DNA, bacteria do not organise their chromosomes on histone octamers, and a nucleosome octamer has two copies each of only four histone types, not eight different kinds. The acidic-protein description reverses the charge logic, since acidic side chains would repel the DNA phosphates.

39. Bacteria are grown for many generations in a medium whose only nitrogen source is the heavy isotope ^{15}N , so that all their DNA is fully heavy. The culture is then transferred to ordinary ^{14}N medium and allowed to complete three further rounds of replication. Assuming semiconservative replication, what fraction of the double-stranded DNA molecules present at the end contain ^{15}N in at least one of their strands?

- A)** $1/16$ **B)** $1/8$ **C)** $1/4$ **D)** $1/2$ **E)** $3/4$

Answer: C

Given: all DNA fully heavy (^{15}N) at the start; 3 further rounds of replication in ^{14}N medium; semiconservative mechanism.

Principle: every original strand persists forever inside exactly one hybrid molecule, while the molecule total doubles each round.

Step 1: molecules containing ^{15}N stay fixed at 2 per starting molecule, one hybrid per original strand, in every round.

Step 2: total molecules after 3 rounds = $2^3 = 8$ per starting molecule.

Step 3: fraction still containing ^{15}N = $2/8 = 1/4$

Result: $1/4$ of the double-stranded molecules contain ^{15}N in at least one strand.

Classic slip: $3/4$ is the fraction that has become entirely light, not the fraction still carrying heavy nitrogen.

40. In a Meselson-Stahl type experiment, bacteria grown in ^{15}N medium are shifted to ^{14}N medium and allowed to replicate for exactly two rounds. Centrifugation then separates the DNA into two distinct bands: one at intermediate density and one fully light, with no heavy band anywhere in the tube. Which conclusion is justified?

- A)** Only the dispersive model is excluded, because the conservative model still predicts an intermediate band.
- B)** Semiconservative replication is excluded, because it predicts a single intermediate band after two rounds.
- C)** Only the conservative model is excluded, because the dispersive model still predicts two separate bands.
- D)** Both the conservative and the dispersive models are excluded, which is the classic evidence for semiconservative replication.
- E)** The experiment is uninterpretable, because no model predicts a light band after only two rounds.

Answer: D

Given: fully heavy start, exactly 2 rounds in light medium; centrifugation shows one intermediate band and one light band, with no heavy band.

Principle: under semiconservative replication each original strand ends up in a hybrid molecule and the molecule count doubles per round.

Step 1: after round 1: 2 molecules per original, both hybrids, one intermediate band.

Step 2: after round 2: 4 molecules per original, made of 2 hybrids (intermediate band) + 2 fully light molecules (light band) + 0 fully heavy.

Step 3: the observed pattern of two bands with no heavy band matches this count exactly.

Step 4: a conservative mechanism would still show a fully heavy band from the untouched original duplex, and a dispersive mechanism would give a single band of uniform intermediate density, so both are excluded.

Result: the observation is the classic evidence for semiconservative replication.

Classic slip: the conservative model also predicts an intermediate band, so an intermediate band alone does not confirm semiconservative replication; the absence of the heavy band is the decisive part.

41. Which statement about the eukaryotic (80S) ribosome is correct?

- A)** One of its subunits is made of protein only and the other of RNA only.
- B)** The complete 80S particle is assembled in the cytoplasm on the rough endoplasmic reticulum.
- C)** Its subunits are assembled in the Golgi apparatus and delivered to the cytoplasm inside transport vesicles.
- D)** It is built from a 50S large subunit and a 30S small subunit, as is the case in all living cells.
- E)** Its large and small subunits are assembled in the nucleolus and are exported separately to the cytoplasm, where they combine only when translation begins.

Answer: E

Ribosomal RNA is transcribed and the 40S and 60S subunits begin assembling in the nucleolus; the mature subunits are exported separately to the cytoplasm and associate only when they meet an mRNA at the start of translation. Each subunit contains both RNA and protein, and the 50S plus 30S combination is the prokaryotic 70S ribosome, not the eukaryotic 80S one.

42. A eukaryotic gene consists of two exons separated by one intron, and that intron contains a stop codon in the reading frame. The DNA of the gene, intron included, is inserted into a bacterium under the control of a bacterial promoter, transcribed and translated. Which outcome is expected?

- A)** The normal protein is produced, with the amino acids encoded by the intron simply inserted between the two exons.
- B)** No polypeptide is produced, because a bacterial ribosome cannot translate a transcript that contains intron sequences.
- C)** A protein longer than the normal one is produced, because the entire intron is translated in addition to both exons.
- D)** A shortened polypeptide is produced, because translation stops at the stop codon inside the intron, before the second exon is ever reached.
- E)** The normal-length protein is produced, because bacteria remove introns from RNA using the same spliceosome as eukaryotes.

Answer: D

Step 1: bacteria lack spliceosomes, so the transcript keeps the intron. Step 2: the bacterial ribosome translates exon 1, continues into the intron and terminates at the in-frame stop codon located there. The result is a truncated polypeptide that lacks the second exon. The idea that the intron's amino acids are simply inserted between the exons fails because translation halts at that stop codon, and bacteria do not remove introns with a eukaryotic spliceosome.

43. Which statement about RNA splicing is correct?

- A)** Splicing removes the exons and joins the introns together to build the mature mRNA that is translated.
- B)** Splicing takes place in the cytoplasm, on the ribosome, immediately before translation begins.
- C)** Splicing alters the nucleotide sequence of the exons so that they come to encode different amino acids.
- D)** Bacterial mRNAs must be spliced before translation, because bacterial genes are normally rich in introns.
- E)** In eukaryotes, introns are removed and exons joined before the mature mRNA leaves the nucleus, whereas most bacterial genes contain no introns and their transcripts are not spliced.

Answer: E

In eukaryotes splicing removes introns and joins exons in the nucleus, before the mature mRNA is exported, whereas bacterial genes are typically intron-free and their transcripts are not spliced. The description that removes exons and joins introns reverses the two elements, and splicing leaves the coding sequence of the exons unchanged.

44. A codon table shows that CAG codes for glutamine and that UAG is a stop codon. In an mRNA molecule, the codon CAG is changed to UAG by a single base substitution. This mutation is best classified as:

- A)** A missense mutation.
- B)** A nonsense mutation.
- C)** A silent mutation.
- D)** An insertion mutation.
- E)** A frameshift mutation.

Answer: B

The substitution CAG to UAG turns a codon specifying glutamine into a stop signal, so translation ends prematurely and a truncated polypeptide results: by definition a nonsense mutation. A missense mutation would specify a different amino acid, a silent one the same amino acid, and a single-base substitution cannot shift the reading frame or count as an insertion.

45. The beginning of an mRNA reads AUG CCU GGU UGA, where UGA is the fourth codon and is a stop signal. Given the code AUG = methionine, CCU = proline, GGU = glycine, ACC = threonine, UGG = tryptophan, UUG = leucine, a mutation inserts one extra nucleotide, A, immediately after the first codon. Which statement is correct?

- A)** Methionine is still the first amino acid, but proline, glycine and every later position change, and the original UGA stop signal is never read because it is split across the new codon boundaries.
- B)** The protein becomes just methionine followed by threonine, because the new codon ACC acts as a stop signal.
- C)** Translation now ends at the position of the former UGA codon, but the amino acids before that point are all unchanged.
- D)** Only one amino acid changes, because the ribosome re-establishes the original reading frame after a single inserted nucleotide.
- E)** The insertion is silent, because adding a single nucleotide leaves each existing codon intact.

Answer: A

Given: mRNA AUG CCU GGU UGA, coding Met, Pro, Gly and then a stop; one extra A inserted immediately after the first codon.

Principle: the ribosome reads fixed triplets from the start codon, so every base after the insertion shifts into a new grouping and the one-base offset persists to the end.

Step 1: regroup the new sequence: AUG | ACC | UGG | UUG, with one base left over.

Step 2: AUG still gives Met; then ACC = Thr, UGG = Trp, UUG = Leu, so Pro and Gly are replaced from position 2 onward.

Step 3: the former stop UGA is split apart: its U and G now sit inside UUG and its A is the leftover tail, so no codon reads UGA any more.

Result: Met first, then Thr, Trp, Leu and beyond: a frameshift that reads straight past the old stop position.

Classic slip: assuming the ribosome re-establishes the original frame after a single inserted base; a one-base offset never heals by itself.

46. A codon table gives: GAA and GAG both code for glutamate, GUA codes for valine, and UAA is a stop codon. In the same gene in two different individuals, single-base substitutions occur at the same codon: in mutant X the codon GAA becomes GAG, and in mutant Y the codon GAA becomes GUA. Which classification is correct?

- A)** X is a frameshift mutation and Y is a missense mutation.
- B)** X is a silent mutation and Y is a missense mutation.
- C)** X is a nonsense mutation and Y is a missense mutation.
- D)** X is a missense mutation and Y is a silent mutation.
- E)** X is a silent mutation and Y is a nonsense mutation.

Answer: B

GAA and GAG both specify glutamate, so mutant X changes no amino acid: a silent mutation. GUA specifies valine, so mutant Y replaces glutamate with a different amino acid: a missense mutation. Neither codon became a stop signal, so classifying either change as nonsense is wrong, and single-base substitutions are not frameshifts.

47. A researcher sets up a PCR containing the DNA template, Taq polymerase and all four dNTPs, but accidentally omits the primers. The thermal cycler still runs its full programme of denaturation, annealing and extension. Which outcome is expected?

- A)** No amplification occurs, because DNA polymerase cannot start a new strand and can only extend from the free 3' end that a paired primer provides.
- B)** The dNTPs anneal to the template in place of the missing primers and allow amplification to proceed.
- C)** The reaction fails because Taq polymerase is destroyed at the annealing temperature of the cycle.
- D)** Amplification occurs but more slowly, because Taq polymerase is able to start synthesis on a single-stranded template without a primer.
- E)** One copy of the template is produced in each cycle, so the DNA accumulates linearly instead of exponentially.

Answer: A

DNA polymerases can only extend an existing strand from a paired 3' end; with no primers in the tube Taq polymerase has nothing to extend, so after each denaturation the template simply re-anneals and no product ever accumulates. Taq is thermostable and survives the cycling, dNTPs cannot anneal in place of primers, and synthesis cannot start on a bare single-stranded template.

48. A PCR tube starts with exactly one double-stranded DNA molecule. The reaction runs three complete cycles, and in every cycle each single strand present serves as the template for one newly synthesized complementary strand. After the third cycle, how many of the double-stranded molecules consist of two newly synthesized strands, with no original template strand at all?

- A)** 0 molecules **B)** 2 molecules **C)** 4 molecules **D)** 6 molecules **E)** 8 molecules

Answer: D

Given: 1 double-stranded starting molecule, 3 complete cycles, and in every cycle each single strand directs the synthesis of one complementary strand.

Principle: double-stranded molecules double each cycle, but the two original strands never leave the pool.

Step 1: after cycle 1: 2 molecules, both hybrids of one original and one new strand.

Step 2: after cycle 2: 4 molecules, of which 2 are hybrids and 2 consist of two new strands.

Step 3: after cycle 3: 8 molecules, and still only 2 hybrids, one per original strand.

Step 4: fully new molecules = $8 - 2 = 6$

Result: 6 molecules consist of two newly synthesized strands.

Classic slip: answering 8 forgets that each original template strand persists forever inside one hybrid molecule.

Section D - Cell cycle, heredity and evolution (Q49-66)

49. A culture of actively dividing fibroblasts is exposed to UV radiation while all cells are in G1, producing numerous thymine dimers. The G1 checkpoint responds normally. Which outcome is expected?

- A)** Cells complete division and enter G0, because DNA repair can occur only after cytokinesis is finished.
- B)** Cells proceed into S phase on schedule, because the G1 checkpoint monitors spindle attachment rather than DNA integrity.
- C)** Cells arrest in G1 and do not enter S phase until the damage is repaired, so damaged DNA is not replicated.
- D)** Cells replicate the damaged DNA normally in S phase and repair it at the G2 checkpoint before mitosis begins.
- E)** Cells arrest in mitosis, because DNA damage is detected only at the metaphase checkpoint.

Answer: C

Step 1: UV produces thymine dimers in the DNA of cells that are all in G1. Step 2: the G1 checkpoint senses DNA damage and halts the cycle, blocking entry into S phase. Step 3: only after the dimers are repaired do the cells proceed to replicate, so damaged DNA is never copied. The G1 checkpoint monitors DNA integrity, not spindle attachment, and postponing repair to G2 would let the damaged templates be replicated first.

50. Which statement about the phases of the eukaryotic cell cycle is correct?

- A)** A cell in the G0 phase actively replicates its DNA in preparation for a future division.
- B)** Cytokinesis, the division of the cytoplasm, is completed during the G2 phase.
- C)** Chromosome condensation and spindle formation take place during the S phase.
- D)** The amount of DNA per cell doubles during mitosis and is halved during the S phase.
- E)** DNA replication takes place during the S phase, so a cell entering mitosis contains twice the amount of DNA it had at the start of interphase.

Answer: E

DNA replication occurs only in S phase, so a cell entering mitosis contains twice the DNA it had at the start of interphase, which is the correct statement. G0 is a quiescent, non-dividing state with no DNA synthesis, chromosome condensation and spindle formation belong to prophase of mitosis, and cytokinesis is part of the M phase, not of G2.

51. During an early mitotic division of a chromosomally normal human embryo ($2n = 46$), the two sister chromatids of chromosome 21 fail to separate in anaphase of one cell. Which outcome is expected in the two daughter cells produced by that division?

- A)** One daughter cell has 45 chromosomes and the other has 45, because both copies of chromosome 21 are lost.
- B)** Both daughter cells have 47 chromosomes, so the whole embryo becomes trisomic for chromosome 21.
- C)** Both daughter cells have 46 chromosomes, because mitotic segregation corrects the failed separation.
- D)** One daughter cell has 47 chromosomes and the other has 45, so the embryo becomes a mosaic of two cell lines.
- E)** One daughter cell has 47 chromosomes and the other has 45, but both lines die so the embryo does not survive.

Answer: D

Given: chromosomally normal embryo cell, $2n = 46$; in one mitotic anaphase the two sister chromatids of chromosome 21 fail to separate.

Principle: in anaphase every separated chromatid moves to one pole or the other; a nondisjunction sends both to one pole and none to the other.

Step 1: a normal pole would receive 46 single-chromatid chromosomes.

Step 2: here one pole receives $46 + 1 = 47$ (both copies of chromosome 21) and the other $46 - 1 = 45$ (none).

Step 3: the daughters therefore hold 47 and 45 chromosomes, both lines keep dividing, and the embryo becomes a mosaic of a trisomic and a monosomic cell line.

Result: one daughter with 47 and one with 45 chromosomes, giving mosaicism.

Classic slip: whole-embryo trisomy would require the error in meiosis; mitosis has no mechanism that corrects a failed segregation.

52. Which of the following statements about meiosis I is false?

- A)** During prophase I, homologous chromosomes pair up in synapsis and may exchange segments by crossing over.
- B)** During anaphase I, sister chromatids separate and move to opposite poles of the cell.
- C)** During metaphase I, paired homologous chromosomes align at the equator, with each pair oriented independently of the others.
- D)** Meiosis I is called the reductional division because it halves the chromosome number, unlike meiosis II.
- E)** At the end of telophase I, each cell contains half the original chromosome number, and its chromosomes are still made of two sister chromatids.

Answer: B

The false statement concerns anaphase I: there, whole homologous chromosomes, each still of two sister chromatids, separate and move to opposite poles, and centromeres split only in anaphase II. The remaining statements match meiosis I correctly: synapsis and crossing over in prophase I, independent orientation of the pairs at the metaphase I equator, half the chromosome number with double-chromatid chromosomes after telophase I, and the name reductional division.

53. Crossing over, a major source of new allele combinations, occurs:

- A)** between non-sister chromatids during anaphase II, when the chromatids separate.
- B)** between homologous chromosomes during the S phase, while the DNA is being copied.
- C)** between non-sister chromatids of homologous chromosomes during prophase I of meiosis.
- D)** between homologous chromosomes during metaphase II, when the pairs align at the equator.
- E)** between the two sister chromatids of the same chromosome during prophase I of meiosis.

Answer: C

Chiasmata form while homologous chromosomes are synapsed in prophase I, and the exchange occurs between non-sister chromatids of the two homologues. An exchange between the two sister chromatids of one chromosome would be genetically invisible, since they carry identical alleles, and neither metaphase II, anaphase II nor S phase is the stage of crossing over.

54. A plant species has a diploid number of $2n = 10$. During anaphase II of meiosis, how many chromosomes move toward each pole, and what does each of them consist of?

- A)** 5 chromosomes, each made of a single chromatid, move to each pole.
- B)** 5 chromosomes, each still made of two sister chromatids, move to each pole.
- C)** 10 chromosomes, each made of two sister chromatids, move to each pole.
- D)** 10 chromosomes, each made of a single chromatid, move to each pole.
- E)** 20 chromatids move as separate chromosomes toward each pole.

Answer: A

Given: diploid number $2n = 10$, stage anaphase II of meiosis.

Principle: meiosis I halves the chromosome number; anaphase II only splits centromeres, it does not change the chromosome count.

Step 1: after meiosis I the cell holds $n = 10 \div 2 = 5$ chromosomes, each still double (two sister chromatids).

Step 2: in anaphase II every centromere splits, so each chromosome becomes single-chromatid.

Step 3: the number moving to each pole stays 5; only the chromatids per chromosome fall from two to one.

Result: 5 chromosomes, each made of a single chromatid, move to each pole.

Classic slip: any count of 10 per pole ignores the halving already accomplished in meiosis I.

55. In a primary spermatocyte with $2n = 12$, a nondisjunction of one homologous pair occurs during meiosis I, and meiosis II then proceeds normally. How many of the four resulting gametes have an abnormal chromosome number?

- A)** 0 **B)** 1 **C)** 2 **D)** 3 **E)** 4

Answer: E

Given: primary spermatocyte with $2n = 12$, so $n = 6$; nondisjunction of one homologous pair in meiosis I; meiosis II normal.

Principle: a meiosis I nondisjunction sends both homologues to one pole, so both cells leaving meiosis I are already unbalanced.

Step 1: end of meiosis I: one cell has $n + 1 = 7$ chromosomes, the other $n - 1 = 5$.

Step 2: meiosis II proceeds normally in both, so each unbalanced cell produces two gametes with its own count: two gametes with 7 and two with 5.

Step 3: abnormal gametes = $2 + 2 = 4$ out of 4

Result: all four gametes have abnormal chromosome numbers.

Classic slip: the pattern of exactly two abnormal gametes is the signature of a meiosis II error, in which one secondary spermatocyte remains normal.

56. A primary spermatocyte ($2n = 24$) completes meiosis I normally. During meiosis II, in one of the two secondary spermatocytes the sister chromatids of one chromosome fail to separate, while the other secondary spermatocyte divides normally. Which set of sperm is produced?

- A)** Two sperm with 12 chromosomes, one sperm with 13 and one sperm with 11 chromosomes.
- B)** Two sperm with 13 chromosomes and two sperm with 11 chromosomes.
- C)** Two sperm with 24 chromosomes and two sperm with 12 chromosomes.
- D)** Four sperm with 12 chromosomes each.
- E)** One sperm with 12 chromosomes, two sperm with 13 and one sperm with 11 chromosomes.

Answer: A

Given: primary spermatocyte with $2n = 24$, so $n = 12$; meiosis I normal; in meiosis II one secondary spermatocyte has a sister-chromatid nondisjunction while the other divides normally.

Principle: a normal meiosis I leaves both secondary spermatocytes balanced at n , and a meiosis II error disturbs only one of them.

Step 1: both secondary spermatocytes enter meiosis II with $n = 12$ double-chromatid chromosomes.

Step 2: the affected one yields one gamete with $n + 1 = 13$ and one with $n - 1 = 11$.

Step 3: the unaffected one yields two normal gametes with 12.

Result: two sperm with 12, one sperm with 13 and one sperm with 11 chromosomes.

Classic slip: the all-abnormal set of two 13s and two 11s would follow from a meiosis I nondisjunction, which unbalances both secondary spermatocytes from the start.

57. A male with $2n = 20$ produces a sperm sample in which a single nondisjunction event occurred in one meiocyte. The four sperm from that meiocyte carry 11, 10, 10 and 9 chromosomes respectively. In which division did the nondisjunction occur, and why?

- A)** Meiosis II, because the failure of sister chromatids to separate makes all four gametes abnormal.
- B)** Meiosis I, because a pair of homologous chromosomes failed to separate and all four sperm are abnormal.
- C)** Meiosis I, because both cells leaving the first division were already aneuploid.
- D)** Either division can be concluded, because both errors produce the same counts of normal and abnormal sperm.
- E)** Meiosis II, because one pair of sister chromatids failed to separate and only one of the two secondary spermatocytes was affected.

Answer: E

Given: $2n = 20$, so the normal sperm count is $n = 10$; the four sperm of one meiocyte carry 11, 10, 10 and 9 chromosomes.

Principle: a meiosis I error unbalances both secondary spermatocytes and so makes all four gametes abnormal; a meiosis II error affects only one secondary spermatocyte and gives two normal plus two aneuploid gametes.

Step 1: two sperm are normal (10, 10) and two are aneuploid (11 and 9).

Step 2: $11 = n + 1$ and $9 = n - 1$ form the matching $+1/-1$ pair produced when sister chromatids fail to separate in one cell only.

Step 3: a pattern of exactly two abnormal sperm is possible only if meiosis I was normal.

Result: the nondisjunction occurred in meiosis II.

Classic slip: a meiosis I homologue failure would give four aneuploid sperm, for example two with 11 and two with 9.

58. Which statement about the origin of genetic variability in sexually reproducing organisms is correct?

- A)** Independent assortment can occur only in cells that have first completed crossing over, since unexchanged pairs always segregate identically.
- B)** Independent assortment arises because each homologous pair takes a random orientation at the metaphase I plate, independently of every other pair.
- C)** Crossing over generates variability by exchanging alleles between the two sister chromatids of a chromosome.
- D)** Random fertilisation adds variability because homologous chromosomes exchange segments just before the gametes fuse.
- E)** Independent assortment occurs during meiosis II, when sister chromatids are distributed at random to the gametes.

Answer: B

Independent assortment arises at metaphase I: each homologous pair attaches to the spindle in a random orientation, independently of every other pair, so maternal and paternal chromosomes are shuffled into the gametes. Crossing over exchanges alleles between non-sister chromatids of homologues, not between the two sister chromatids of a chromosome, and it is not a prerequisite for independent assortment, which occurs even in its absence. Random fertilisation adds variability by combining random gametes, not by any exchange of chromosome segments.

59. In an animal species with $2n = 12$, how many genetically different chromosome combinations can the gametes acquire through independent assortment alone, ignoring crossing over?

- A)** 12 **B)** 24 **C)** 36 **D)** 64 **E)** 4096

Answer: D

Given: $2n = 12$; variability from independent assortment alone, ignoring crossing over.

Principle: each of the n homologous pairs orients independently at metaphase I, so the number of gamete chromosome combinations is 2^n .

Step 1: number of homologous pairs = $12 \div 2 = 6$, so $n = 6$

Step 2: combinations = $2^6 = 64$

Result: 64 genetically different chromosome combinations can be produced.

Classic slip: raising 2 to the diploid number 12 gives 4096, which wrongly counts every single chromosome as an independent unit, and 24 is merely the diploid count itself.

60. In humans the M and N blood-group alleles are codominant. A couple are both of type MN. Which phenotype ratio is expected among their children?

- A)** 1 child of type M for every 2 children of type MN and 1 child of type N
- B)** 1 child of type M for every 1 child of type MN and 1 child of type N
- C)** 2 children of type MN for every 1 child of type M and 1 child of type N
- D)** 3 children of type M for every 1 child of type N
- E)** 9 children of type M for every 3 children of type MN and 4 children of type N

Answer: A

Given: both parents are MN, with the codominant alleles M and N.

Principle: each parent produces M and N gametes in equal numbers, so Punnett-square counting gives the genotype ratio, and codominance displays every genotype as its own phenotype.

Step 1: gametes: M or N from each parent.

Step 2: offspring combinations: MM once, MN twice, NN once, i.e. 1 MM : 2 MN : 1 NN.

Step 3: codominance makes MM read as type M, MN as type MN and NN as type N, all three visibly distinct.

Result: phenotype ratio 1 type M : 2 type MN : 1 type N.

Classic slip: a 3:1 ratio would require M to be dominant over N, which codominance rules out.

61. Which of the following statements about Mendelian crosses is false?

- A)** A cross between two heterozygous parents for a codominant trait gives a 1:2:1 phenotypic ratio.
- B)** A testcross between a heterozygous dominant individual and a homozygous recessive partner gives a 1:1 phenotypic ratio.
- C)** A cross between a homozygous dominant individual and a homozygous recessive partner gives a 1:1 phenotypic ratio.
- D)** A monohybrid cross between two heterozygous parents gives a 3:1 ratio of dominant to recessive phenotypes.
- E)** A dihybrid cross between two individuals heterozygous for both genes gives a 9:3:3:1 phenotypic ratio when the genes assort independently.

Answer: C

A homozygous dominant parent crossed with a homozygous recessive one yields only heterozygous offspring, all showing the dominant phenotype, so the claim of a 1:1 ratio from that cross is false. The 1:1 ratio is the signature of the testcross, a heterozygote against a homozygous recessive, and the 3:1 monohybrid, 9:3:3:1 dihybrid and codominant 1:2:1 statements are all correct.

62. Two pea plants heterozygous for seed shape and for seed colour ($AaBb \times AaBb$) are crossed, and the two genes assort independently. How many different genotypes are expected among the offspring?

- A)** 4 **B)** 6 **C)** 9 **D)** 12 **E)** 16

Answer: C

Given: the cross $AaBb \times AaBb$ with independent assortment; the question asks for distinct genotypes.

Principle: with independent assortment, genotype counts multiply across loci (product rule).

Step 1: at the A locus, $Aa \times Aa$ gives 3 genotypes: AA, Aa, aa.

Step 2: at the B locus, $Bb \times Bb$ likewise gives 3 genotypes: BB, Bb, bb.

Step 3: total genotypes = $3 \times 3 = 9$

Result: 9 different genotypes among the offspring.

Classic slip: 16 counts the Punnett squares and 4 counts the phenotype classes; neither is the number of distinct genotypes.

63. In a cross $AaBb \times aaBb$, with the two genes assorting independently, what is expected among the offspring?

- A)** Four different phenotypes, with three sixteenths of the offspring showing both recessive traits.
- B)** Four different phenotypes, with one eighth of the offspring showing both recessive traits ($aabb$).
- C)** Two different phenotypes, with one eighth of the offspring showing both recessive traits.
- D)** Three different phenotypes, with one quarter of the offspring showing both recessive traits.
- E)** Four different phenotypes, with one sixteenth of the offspring showing both recessive traits.

Answer: B

Given: the cross $AaBb \times aaBb$ with independent assortment; the question asks for the phenotype classes and the fraction showing both recessive traits ($aabb$).

Principle: treat one locus at a time, then multiply the fractions (product rule).

Step 1: A locus, $Aa \times aa$: $1/2$ Aa (dominant phenotype) and $1/2$ aa , so $P(A_) = 1/2$ and $P(aa) = 1/2$

Step 2: B locus, $Bb \times Bb$: $3/4$ $B_$ and $1/4$ bb .

Step 3: multiply: $A_B_ = 1/2 \times 3/4 = 3/8$; $aaB_ = 1/2 \times 3/4 = 3/8$; $A_bb = 1/2 \times 1/4 = 1/8$; $aabb = 1/2 \times 1/4 = 1/8$

Step 4: distinct phenotype classes = 4

Result: four phenotypes, with $1/8$ of the offspring $aabb$.

Classic slip: $1/16$ belongs to the self-cross $AaBb \times AaBb$, not to this cross.

64. In a population of 500 people at Hardy-Weinberg equilibrium, 20 people show an autosomal recessive condition (genotype aa). How many people are expected to be heterozygous carriers?

- A)** 80 **B)** 100 **C)** 160 **D)** 200 **E)** 320

Answer: C

Given: population of 500 at Hardy-Weinberg equilibrium; 20 people affected with the autosomal recessive genotype aa ; the question asks for heterozygous carriers.

Principle: Hardy-Weinberg: q^2 is the frequency of aa , $p + q = 1$, and the carrier frequency is $2pq$.

Step 1: $q^2 = 20/500 = 0.04$

Step 2: $q = \sqrt{0.04} = 0.2$

Step 3: $p = 1 - 0.2 = 0.8$

Step 4: carrier frequency = $2pq = 2 \times 0.8 \times 0.2 = 0.32$

Step 5: carriers = $0.32 \times 500 = 160$ people

Result: 160 heterozygous carriers.

Classic slip: 80 uses $pq = 0.16$ and drops the factor of 2 in $2pq$.

65. A population is in Hardy-Weinberg equilibrium, and 9 percent of its members show a phenotype caused by a recessive allele. What is the frequency of the dominant allele in this population?

- A)** 0.3 **B)** 0.49 **C)** 0.51 **D)** 0.7 **E)** 0.9

Answer: D

Given: population in Hardy-Weinberg equilibrium; 9 percent show the recessive phenotype; the question asks for the frequency of the dominant allele.

Principle: the recessive phenotype reveals q^2 , and $p = 1 - q$

Step 1: $q^2 = 0.09$

Step 2: $q = \sqrt{0.09} = 0.3$

Step 3: $p = 1 - 0.3 = 0.7$

Result: the dominant allele has frequency 0.7.

Classic slip: 0.49 is p^2 , the frequency of homozygous dominant individuals, not the allele frequency itself.

66. Which statement about evolution and the history of evolutionary thought is correct?

- A)** Pasteur's experiment demonstrated that new species arise through the inheritance of characteristics acquired during an organism's life.
- B)** After a bottleneck, the surviving population keeps the allele frequencies of the original population, because which individuals survive is random.
- C)** A population bottleneck increases genetic diversity, because only the best-adapted individuals survive the crisis.
- D)** The bottleneck effect generates new alleles, because mutation rates rise when a population becomes small.
- E)** Pasteur's swan-necked flask experiment showed that microorganisms appear in nutrient broth only when the broth is exposed to airborne microbes, refuting spontaneous generation.

Answer: E

Pasteur's swan-necked flasks admitted air but trapped microbes and dust in the curved neck; the broth stayed sterile until the neck was broken, showing that microorganisms arise only from existing microbes and refuting spontaneous generation. A bottleneck is a random sampling event: it loses alleles and lowers genetic diversity rather than raising it, produces no new alleles, and does not preserve the original allele frequencies.

Section E - Animal tissues (Q67-74)

67. Which option pairs every site with the epithelium that normally lines it?

- A) Alveolus: simple squamous; trachea: ciliated pseudostratified columnar; oesophagus: transitional; small intestine: simple columnar
- B) Alveolus: simple squamous; trachea: stratified squamous; oesophagus: stratified squamous; small intestine: simple columnar
- C) Alveolus: simple cuboidal; trachea: ciliated pseudostratified columnar; oesophagus: stratified squamous; small intestine: simple columnar
- D) Alveolus: simple squamous; trachea: ciliated pseudostratified columnar; oesophagus: stratified squamous; small intestine: simple cuboidal
- E) Alveolus: simple squamous; trachea: ciliated pseudostratified columnar; oesophagus: stratified squamous; small intestine: simple columnar

Answer: E

Alveoli are lined by simple squamous epithelium for short diffusion paths, the trachea by ciliated pseudostratified columnar epithelium carrying the mucus escalator, the oesophagus by stratified squamous epithelium against abrasion, and the small intestine by simple columnar epithelium for absorption. The cuboidal alveolus claim fits only the surfactant-secreting type II cells, not the exchange surface itself, and transitional epithelium belongs to the urinary tract, never the oesophagus.

68. Which of the following statements about epithelial tissue is false?

- A) An epithelium is avascular and takes its nourishment by diffusion from capillaries in the underlying connective tissue.
- B) Pseudostratified columnar epithelium is built of several true layers, because only some of its cells reach the free surface.
- C) Epithelia are classified by the number of cell layers and by the shape of the cells at the free surface.
- D) In stratified epithelia the tissue is named for the shape of the cells at the apical surface, not the basal layer.
- E) A gland that develops as an invagination of an epithelium and keeps a duct is still classified as epithelial tissue.

Answer: B

Pseudostratified columnar epithelium is a single layer: every cell rests on the basement membrane, and only the staggered nuclei together with some cells not reaching the free surface create the illusion of layers, so the note calling it several true layers is the false one. The classification rule (layer number plus apical cell shape), the epithelial nature of ducted glands, and avascular epithelia nourished by diffusion from the connective tissue beneath are all correctly stated.

69. A light micrograph shows a single layer of tall cells, all resting on the same basement membrane. The nuclei lie at several different heights, giving a layered look, and the free surface carries motile projections. This epithelium is:

- A) stratified squamous epithelium lining the vagina
- B) simple columnar epithelium with a microvillus border lining the small intestine
- C) stratified columnar epithelium lining the oesophagus
- D) ciliated pseudostratified columnar epithelium lining the trachea
- E) transitional epithelium lining the urinary bladder

Answer: D

Nuclei at several heights with every cell still resting on the same basement membrane define pseudostratified columnar epithelium, and the motile surface projections are cilia. That combination is the lining of the trachea. Genuinely stratified columnar tissue would have cells stacked in true layers, transitional epithelium lines the urinary bladder, and a microvillus border is non-motile.

70. Oxygen travels from the air inside an alveolus into the blood by crossing the alveolar wall and then the capillary wall. Which statement about these exchange surfaces is correct?

- A)** Both walls are simple squamous epithelium, the thinnest arrangement possible, which keeps the diffusion path short.
- B)** The alveolar wall is ciliated pseudostratified columnar epithelium, a direct continuation of the airway lining.
- C)** The alveolar wall is stratified squamous epithelium, because the moving air continually abrades it.
- D)** The capillary wall is ciliated epithelium, and the cilia propel the blood along the vessel.
- E)** Both walls are simple cuboidal epithelium, whose extra cytoplasm pumps the gases across actively.

Answer: A

Both the alveolar wall (type I alveolar cells) and the capillary wall (endothelium) are simple squamous epithelium, the flattest lining possible, so the air-to-blood diffusion barrier is kept minimal. Gases cross passively and need no pumping cytoplasm, the ciliated pseudostratified lining belongs to the conducting airways rather than the alveoli, and blood is driven by the heart, not by cilia.

71. A toxin selectively destroys the brush border of the cells lining the proximal convoluted tubule, leaving the rest of the nephron intact. Which outcome is expected?

- A)** Urine volume falls to nearly zero, because most of the filtrate is produced in the proximal tubule.
- B)** Glomerular filtration stops, because the podocytes of the capsule depend on the same brush border.
- C)** The collecting duct takes over glucose reabsorption, so the composition of the urine stays normal.
- D)** Nothing changes, because microvilli are a feature of the small intestine and not of the kidney.
- E)** Glucose appears in the urine, because the loss of microvilli removes much of the reabsorptive surface and transport capacity of the tubule.

Answer: E

Step 1: proximal tubule cells bear a microvillus brush border carrying the co-transporters that reabsorb essentially all filtered glucose. Step 2: destroying the border removes that surface and transport capacity, so glucose escapes into the urine. Glomerular filtration continues unchanged, the collecting duct cannot take over glucose reabsorption, and the brush border is a normal feature of the kidney proximal tubule, not exclusive to the intestine.

72. Which nephron segment is correctly matched to its lining epithelium and to the reason that epithelium suits it?

- A)** Thin limb of the loop of Henle: simple squamous epithelium, a wall so thin that water and solutes cross it passively with little resistance.
- B)** Proximal convoluted tubule: simple squamous epithelium, whose flat cells make active pumping of glucose faster.
- C)** Collecting duct: transitional epithelium, so that the duct can stretch as urine volume rises.
- D)** Distal convoluted tubule: ciliated columnar epithelium, whose cilia keep the filtrate well mixed.
- E)** Thin limb of the loop of Henle: stratified squamous epithelium, a many-layered wall that resists abrasion by the filtrate.

Answer: A

The thin limb of the loop of Henle is lined by simple squamous epithelium, whose extreme thinness suits the passive movement of water and solutes on which the counter-current system depends. The proximal tubule is simple cuboidal with a brush border for bulk active reabsorption, transitional epithelium lines the renal pelvis, ureters and bladder rather than any nephron segment, and nephron walls are neither ciliated nor stratified.

73. A student wrote the following notes on the surface specializations of epithelial cells. Which note is correct?

- A)** Goblet cells release a thin, enzyme-rich secretion that chemically breaks down particles trapped on the epithelium.
- B)** Tight junctions anchor the intermediate filaments of neighbouring cells to each other, spreading mechanical stress across the sheet.
- C)** The stereocilia of the epididymis beat rhythmically to drive sperm towards the vas deferens.
- D)** Microvilli are non-motile finger-like folds supported by bundles of actin filaments, and they enlarge the absorptive surface of the cell.
- E)** Cilia contain an actin core and bend through dynein-powered sliding to sweep fluid along the epithelial surface.

Answer: D

Microvilli are non-motile finger-like folds whose cores are bundles of actin filaments, and their job is enlarging the apical surface for absorption, as in the intestinal brush border, which is the correct note. Cilia are built on a microtubule axoneme driven by dynein, not on actin; goblet cells secrete mucus, not enzymes; anchoring intermediate filaments is the job of desmosomes, not tight junctions; and epididymal stereocilia do not beat.

74. In a hereditary disorder the dynein arms of cilia are absent, so cilia cannot beat. Which consequence is expected?

- A)** Mucus clearance is unaffected, because the mucus blanket is dragged along by microvilli rather than cilia.
- B)** Mucus accumulates in the airways and transport of the egg along the oviduct is impaired.
- C)** Nutrient absorption in the small intestine largely fails, because microvilli use the same dynein motor as cilia.
- D)** Sperm swim normally, because a flagellum is built on a different internal framework from a cilium.
- E)** Only the airways are affected, because ciliated cells occur nowhere else in the human body.

Answer: B

Step 1: motile cilia and the sperm flagellum share the same microtubule axoneme driven by dynein arms. Step 2: without dynein, airway cilia cannot sweep the mucus blanket, oviduct cilia cannot transport the egg, and sperm cannot swim. Microvilli have actin cores and no dynein, so nutrient absorption is unaffected, and ciliated epithelia occur in many organs, not only in the airways.

Section F - Cardiovascular, respiratory and blood physiology (Q75-91)

75. A patient's mitral (bicuspid) valve no longer closes completely. What happens during ventricular systole?

- A)** Blood regurgitates from the right ventricle back into the right atrium, so venous return falls.
- B)** Blood flows from the left atrium into the left ventricle more rapidly during systole.
- C)** Blood regurgitates from the left ventricle back into the left atrium, so the stroke volume entering the aorta is reduced.
- D)** Blood regurgitates from the aorta back into the left ventricle, so aortic pressure rises.
- E)** The valve cannot open, so the left ventricle fills completely during diastole.

Answer: C

Step 1: the mitral valve sits between the left atrium and the left ventricle and must stay closed while that ventricle contracts. Step 2: incompetence lets part of the ejected blood leak back into the left atrium during systole. Step 3: the forward stroke volume entering the aorta is reduced. Backflow from the aorta describes a leaking aortic valve, which leaks during diastole, and the right-sided chambers are not involved at all.

76. The aortic semilunar valve of a patient becomes incompetent, so that it fails to close after ventricular systole. Which combination of effects during ventricular diastole follows?

- A)** Blood flows from the left atrium into the aorta and ventricular filling is prevented.
- B)** Blood flows from the pulmonary artery into the left ventricle and arterial pressure rises.
- C)** Blood leaks from the aorta back into the left atrium, so atrial pressure rises.
- D)** Blood leaks from the aorta back into the left ventricle, so aortic pressure falls steeply during diastole while the ventricle receives an extra volume of blood.
- E)** Blood stays in the aorta because the valve cannot reopen during diastole, so arterial pressure is unchanged.

Answer: D

Step 1: the aortic valve must hold up the column of blood in the aorta between beats. Step 2: if it leaks, blood runs from the aorta back into the relaxing left ventricle during diastole. Step 3: aortic pressure falls steeply in diastole and the ventricle ends up overfilled with the extra volume before the next beat. The aorta and left atrium are not directly connected, since the mitral valve is the one separating those two chambers, and the valve lesion cannot leave arterial pressure unchanged.

77. Which pairing of a heart valve with its position and its type is correct?

- A)** The bicuspid (mitral) valve is an atrioventricular valve between the left atrium and the left ventricle.
- B)** The bicuspid (mitral) valve lies between the right atrium and the right ventricle.
- C)** The aortic valve is an atrioventricular valve between the left ventricle and the left atrium.
- D)** The tricuspid valve is a semilunar valve between the right ventricle and the pulmonary artery.
- E)** The pulmonary valve is an atrioventricular valve between the right atrium and the right ventricle.

Answer: A

The two atrioventricular valves lie between atrium and ventricle: tricuspid on the right side and bicuspid (mitral) on the left, while the aortic and pulmonary valves form the semilunar pair at the exits of the ventricles. So the correct pairing is the bicuspid valve as an atrioventricular valve between left atrium and left ventricle. Calling the tricuspid a semilunar valve between right ventricle and pulmonary artery misplaces that valve and assigns it the wrong class.

78. A drop of blood enters the heart through the superior vena cava. On its way to the aorta it must pass the following structures: pulmonary artery, pulmonary vein, right ventricle, left atrium, left ventricle. In which order does it meet them?

- A)** Pulmonary artery, right ventricle, pulmonary vein, left ventricle, left atrium.
- B)** Left atrium, left ventricle, pulmonary artery, pulmonary vein, right ventricle.
- C)** Right ventricle, pulmonary artery, pulmonary vein, left atrium, left ventricle.
- D)** Right ventricle, pulmonary vein, pulmonary artery, left ventricle, left atrium.
- E)** Right ventricle, pulmonary artery, left atrium, pulmonary vein, left ventricle.

Answer: C

Step 1: from the superior vena cava the blood enters the right atrium and then the right ventricle. Step 2: it leaves through the pulmonary artery to the lungs. Step 3: it returns by the pulmonary veins to the left atrium and passes into the left ventricle, whose outflow is the aorta. The order is right ventricle, pulmonary artery, pulmonary vein, left atrium, left ventricle. The sequence that meets the pulmonary vein before the pulmonary artery reverses the actual direction of pulmonary flow.

79. A red blood cell leaves the right ventricle and is followed until it re-enters the heart. Which option gives the correct route, with the oxygenation status of the blood at each named stage?

- A)** Pulmonary artery (deoxygenated), lung capillaries, pulmonary vein (oxygenated), right atrium.
- B)** Pulmonary artery (deoxygenated), lung capillaries, pulmonary vein (oxygenated), left atrium.
- C)** Pulmonary vein (deoxygenated), lung capillaries, pulmonary artery (oxygenated), left atrium.
- D)** Pulmonary artery (oxygenated), lung capillaries, pulmonary vein (deoxygenated), right atrium.
- E)** Pulmonary artery (oxygenated), lung capillaries, pulmonary vein (oxygenated), left atrium.

Answer: B

The right ventricle pumps deoxygenated blood into the pulmonary artery; the blood is oxygenated in the lung capillaries and returns through the pulmonary veins to the left atrium. These vessels are the two famous exceptions to the usual rules: a pulmonary artery carrying deoxygenated blood and pulmonary veins carrying oxygenated blood. The route ending at the right atrium gets the vessels right but delivers the blood to the wrong side of the heart.

80. Which statement about the human double circulation is correct?

- A)** The right ventricle has the thickest muscular wall because it serves all the organs of the body.
- B)** The left ventricle pumps blood to the lungs, where the pressure must be high to push blood through the alveolar capillaries.
- C)** Systemic arteries and pulmonary arteries both carry oxygenated blood because all arteries leave the heart.
- D)** Pulmonary veins carry deoxygenated blood because every vein in the body carries deoxygenated blood.
- E)** The right side of the heart pumps blood through the short pulmonary circuit, so blood in the pulmonary arteries travels at a lower pressure than blood in the aorta.

Answer: E

The pulmonary circuit is short and low-resistance, served by the thinner-walled right ventricle, so blood in the pulmonary arteries travels at far lower pressure than blood in the aorta, which is the correct statement. The thickest wall belongs to the left ventricle serving the whole body, pulmonary arteries carry deoxygenated and pulmonary veins oxygenated blood, and the left ventricle supplies the systemic circuit, not the lungs.

81. At the start of a quiet inhalation the diaphragm contracts. Which chain of events follows?

- A)** The diaphragm relaxes, thoracic volume decreases, alveolar pressure falls below atmospheric pressure, and air flows inward.
- B)** The diaphragm domes upward, thoracic volume increases, alveolar pressure rises above atmospheric pressure, and air flows inward.
- C)** The diaphragm flattens, thoracic volume decreases, alveolar pressure rises above atmospheric pressure, and air flows inward.
- D)** The diaphragm flattens, thoracic volume increases, alveolar pressure rises above atmospheric pressure, and air flows outward.
- E)** The diaphragm flattens, thoracic volume increases, pressure inside the alveoli falls below atmospheric pressure, and air flows inward.

Answer: E

Step 1: the contracting diaphragm flattens and enlarges the thoracic cavity. Step 2: by the inverse volume-pressure relation, alveolar pressure falls below atmospheric. Step 3: air flows down this gradient into the lungs. Any chain in which the thoracic volume decreases, or in which alveolar pressure rises while air still flows inward, breaks the pressure-flow logic, and a relaxing diaphragm belongs to exhalation.

82. A person blows out a candle forcefully. Which combination of events occurs during this forced exhalation?

- A)** The abdominal muscles and the internal intercostal muscles contract, the relaxed diaphragm is pushed upward, thoracic volume decreases, and alveolar pressure rises well above atmospheric pressure.
- B)** The external intercostal muscles lift the ribs while the diaphragm flattens, so thoracic volume increases.
- C)** The diaphragm contracts and alveolar pressure falls below atmospheric pressure.
- D)** The diaphragm relaxes while the ribs are pulled upward and outward, so thoracic volume increases.
- E)** Only the diaphragm relaxes, exactly as in quiet breathing at rest, and no other muscles contribute, so alveolar pressure rises only slightly.

Answer: A

Forced exhalation is active: the abdominal muscles push the relaxed diaphragm upward and the internal intercostals pull the ribs down and in, so thoracic volume falls below its resting level, alveolar pressure rises well above atmospheric, and air is driven out. Pulling the ribs upward and outward while the diaphragm flattens is the pattern of inhalation, and quiet exhalation at rest involves no extra muscles but is also far weaker.

83. Which statement about the pleural membranes is correct?

- A)** The pleural membranes secrete the mucus that moistens the alveoli for gas exchange.
- B)** The pleural cavity normally holds a thin film of serous fluid, so the two layers slide smoothly over each other and the lungs follow the movements of the chest wall.
- C)** The pleural cavity is a large air-filled space that lets the lungs float freely.
- D)** The pleural cavity contains cartilage rings that keep the bronchi open during breathing.
- E)** The visceral pleura lines the inside of the chest wall, and the parietal pleura covers the lung surface.

Answer: B

The parietal pleura lines the inside of the chest wall and the visceral pleura covers the lung surface; between them the pleural cavity holds only a thin film of serous fluid under negative pressure, which lubricates the surfaces and couples each lung to the chest wall. The pairing that puts the visceral pleura on the chest wall has the two layers exactly backwards, the cavity is not air-filled, the pleura secretes serous fluid rather than mucus, and cartilage rings belong to the bronchi.

84. A stab wound lets air enter the left pleural cavity (pneumothorax). What immediate effect does this have on the left lung?

- A)** Nothing changes in the lung, because each side's diaphragm contracts independently of the pleura.
- B)** The left lung stiffens and expands, because serous fluid in the cavity is replaced by air.
- C)** The left lung inflates further, because the extra air increases the pressure gradient into the alveoli.
- D)** The left lung recoils and collapses, because intrapleural pressure rises toward atmospheric pressure and the coupling between chest wall and lung is lost.
- E)** The trachea seals the left main bronchus, so airflow into the left lung simply stops.

Answer: D

Step 1: ventilation depends on the negative pressure of the pleural cavity holding each lung stretched against the chest wall. Step 2: air entering the cavity raises intrapleural pressure toward atmospheric and abolishes that coupling. Step 3: the elastic left lung recoils and collapses. No reflex seals a bronchus, and the injury removes the ventilatory pressure gradient rather than strengthening it.

85. In human blood, which option gives the approximate shares of carbon dioxide transported as bicarbonate ions, bound to hemoglobin, and dissolved in plasma, respectively?

- A)** About 50%, about 43%, about 7%.
- B)** About 23%, about 70%, about 7%.
- C)** About 70%, about 23%, about 7%.
- D)** About 97%, about 3%, about 0%.
- E)** About 70%, about 7%, about 23%.

Answer: C

Carbon dioxide travels about 70% as bicarbonate ions, about 23% bound to the globin of haemoglobin as carbaminohaemoglobin, and only about 7% dissolved in plasma. The figures 97% and 3% belong to oxygen transport, and swapping the bicarbonate and haemoglobin-bound shares is the classic slip.

86. Oxygen and carbon dioxide are both carried with the help of hemoglobin. Which statement about their binding sites on hemoglobin is correct?

- A)** Oxygen binds to the globin chains, while carbon dioxide binds to the cell membrane of the red blood cell.
- B)** Oxygen binds to the heme iron, while carbon dioxide is carried entirely dissolved in plasma with no binding to hemoglobin.
- C)** Both gases bind to the iron of the heme group and compete for the same site.
- D)** Oxygen binds to the globin chains, while carbon dioxide binds to the iron of the heme group.
- E)** Oxygen binds to the iron of the heme group, while carbon dioxide binds to amino groups of the globin polypeptide chains.

Answer: E

Oxygen binds the iron atom of the heme group, forming oxyhaemoglobin, while carbon dioxide binds to amino groups of the globin polypeptide chains, forming carbaminohaemoglobin. The two gases therefore occupy different parts of the molecule and can be carried simultaneously, so the claim of a shared heme-iron site is wrong. Nor is carbon dioxide carried only in solution, since roughly a quarter of it rides on haemoglobin.

87. Approximately what percentage of the oxygen transported in systemic arterial blood is carried bound to hemoglobin as oxyhemoglobin?

- A)** 3% **B)** 7% **C)** 23% **D)** 50% **E)** 97%

Answer: E

About 97% of the oxygen in systemic arterial blood is bound to haemoglobin as oxyhaemoglobin, and only about 3% dissolves in the plasma. The figures 7% and 23% describe the dissolved and haemoglobin-bound shares of carbon dioxide transport, not of oxygen.

88. A patient has a severely reduced hemoglobin concentration, yet the partial pressure of oxygen in her arterial blood is normal. Which statement explains why her tissues still receive too little oxygen?

- A)** The dissolved fraction rises until it equals the normal total oxygen content of the blood.
- B)** Dissolved oxygen in plasma normally supplies about 70% of the tissues' demand, so small changes matter greatly.
- C)** Her alveolar walls have become too thick for oxygen to diffuse across at a normal partial pressure.
- D)** About 97% of the oxygen in arterial blood is bound to hemoglobin, so with less hemoglobin the total oxygen content of the blood falls even though its partial pressure is normal.
- E)** With less hemoglobin, oxygen binds more tightly to the remaining molecules and therefore cannot be released at the tissues.

Answer: D

Step 1: the arterial partial pressure fixes how far the available haemoglobin saturates, but the total oxygen content of the blood depends on how much haemoglobin there is to load. Step 2: since about 97% of arterial oxygen travels on haemoglobin, less haemoglobin cuts total delivery even though partial pressure and the small dissolved share stay normal. Thickened alveolar walls are not the lesion in anaemia, dissolved oxygen covers only a few percent of demand, and a reduced haemoglobin pool does not make the remaining molecules hold oxygen more tightly.

89. Emphysema progressively destroys the walls between the pulmonary alveoli. Which direct consequence for gas exchange follows?

- A)** The bronchioles constrict by smooth muscle spasm, so air cannot reach the alveoli.
- B)** The pleural cavity fills with mucus, so the lungs cannot slide along the chest wall.
- C)** The total surface area available for gas exchange decreases, so oxygen uptake per breath falls.
- D)** The amount of hemoglobin in the blood falls, so oxygen carriage drops.
- E)** The diaphragm can no longer contract, so thoracic volume cannot increase.

Answer: C

The alveolar walls are themselves the gas-exchange surface, so emphysema, by merging alveoli into fewer, larger sacs, directly cuts the diffusion area and oxygen uptake per breath falls. Bronchiolar smooth-muscle spasm defines asthma, a falling haemoglobin count defines anaemia, and neither diaphragm contraction nor pleural fluid is touched by alveolar destruction.

90. Which pairing of a condition with the structure primarily affected is incorrect?

- A)** Asthma: destruction of the walls between the alveoli.
- B)** Myocardial infarction: a coronary artery.
- C)** Chronic bronchitis: the mucus-producing glands of the bronchi.
- D)** Varicose veins: incompetent valves in the veins of the legs.
- E)** Pleurisy: the pleural membranes.

Answer: A

Asthma is a reversible narrowing of the airways caused by bronchiolar smooth-muscle contraction together with inflammation; destruction of the walls between the alveoli is the hallmark of emphysema, so that condition-to-structure pairing is the incorrect one. Pleurisy, myocardial infarction, varicose veins and chronic bronchitis are matched correctly to the pleural membranes, a coronary artery, incompetent leg-vein valves and the bronchial mucus glands respectively.

91. Fatty plaques narrow a patient's coronary arteries. Why is the heart muscle endangered especially during physical exertion?

- A)** Because exertion diverts all coronary blood flow to the skeletal muscles, leaving the myocardium without perfusion even in healthy people.
- B)** Because the coronary arteries supply the myocardium itself, and exertion raises the heart's oxygen demand faster than blood flow through the narrowed vessels can increase.
- C)** Because the coronary arteries supply the lungs, and exertion raises pulmonary blood flow above what the narrowed vessels can carry.
- D)** Because coronary arteries fill with blood during ventricular systole, and the higher heart rate of exertion shortens systole below what filling needs.
- E)** Because the myocardium switches to anaerobic fermentation during exertion, and the narrowed arteries cannot deliver the extra glucose this requires.

Answer: B

The coronary arteries branch from the aorta and feed the myocardium itself. During exertion the heart's oxygen demand climbs steeply, and vessels narrowed by fatty plaques cannot raise their flow enough, so the muscle becomes ischemic. Coronary filling occurs mainly during diastole, not systole, the coronary circulation never supplies the lungs, and the myocardium does not adopt anaerobic fermentation as its exercise strategy.

Section G - Digestive, endocrine, renal and nervous systems (Q92-112)

92. In the small intestine, the enzyme enterokinase converts pancreatic trypsinogen into active trypsin, and trypsin then activates the other pancreatic proenzymes. A drug that fully inhibits enterokinase is given. Which outcome is expected?

- A)** Protein digestion in the intestine is strongly impaired, because trypsin and the proenzymes it activates stay inactive, while fat and starch digestion continue, since lipase and amylase are secreted already active.
- B)** Starch digestion stops, because pancreatic amylase is released as an inactive zymogen that only trypsin can switch on.
- C)** Protein digestion ceases entirely, because pepsinogen can no longer be activated without enterokinase.
- D)** All digestion of food in the intestine stops, because every pancreatic enzyme is released as an inactive precursor.
- E)** Fat digestion falls to almost zero, because trypsin must first convert prolipase into active lipase.

Answer: A

Step 1: only the proteolytic pancreatic enzymes travel as inactive zymogens. Step 2: without enterokinase no trypsin is formed, so trypsinogen, chymotrypsinogen and procarboxypeptidase all remain inactive and protein digestion is strongly impaired. Step 3: pancreatic lipase and amylase are secreted already active, so fat and starch digestion continue. Pepsinogen is activated by stomach acid rather than enterokinase, so gastric pepsin is untouched.

93. Which statement about bile in human digestion is correct?

- A)** Bile is produced by the pancreas and stored by the liver until a fatty meal arrives.
- B)** Bile is produced by the liver and digests triglycerides into fatty acids and glycerol by means of its own enzymes.
- C)** Bile is produced by the liver and its main digestive role is the breakdown of nucleic acids into nucleotides.
- D)** Bile is produced by the liver and emulsifies fat into fine droplets, enlarging the surface area exposed to lipase, but it hydrolyses no chemical bonds.
- E)** Bile is produced by the gallbladder, which synthesises it anew each time a fatty meal reaches the duodenum.

Answer: D

Bile is produced by the liver, stored and concentrated in the gallbladder, and released into the duodenum; its bile salts emulsify fat into fine droplets, enlarging the surface exposed to lipase, and bile contains no digestive enzymes, so it hydrolyses nothing. The gallbladder does not synthesise bile afresh, and nucleic-acid digestion belongs to the pancreatic nucleases.

94. A drug almost completely abolishes hydrochloric acid secretion by the stomach. Which outcome follows most directly?

- A)** Bile release stops, because the gallbladder contracts only in response to acid produced by the stomach.
- B)** Gastric protein digestion is unaffected, because pepsinogen activates itself independently of pH once the food is churned.
- C)** Activation of pepsinogen into pepsin is largely prevented, so gastric protein digestion slows, while digestion of protein can still continue in the intestine through pancreatic proteases.
- D)** Gastric fat digestion stops completely, because gastric lipase is active only at pH 2 or below.
- E)** The stomach stops secreting pepsinogen altogether, because pepsinogen release is triggered by acid itself.

Answer: C

Step 1: gastric acid both converts pepsinogen into pepsin and supplies the low pH at which pepsin works, so abolishing acid largely prevents activation and slows gastric protein digestion. Step 2: protein digestion can still continue in the small intestine through the pancreatic proteases. Pepsinogen secretion itself does not require acid, gastric lipase works at the moderately acidic gastric pH rather than only at pH 2, and gallbladder contraction is driven mainly by cholecystokinin, not by gastric acid.

95. Rennin (chymosin), secreted by the gastric mucosa of young mammals, turns liquid milk into a solid curd so that it stays longer in the stomach. The substance on which rennin acts is:

- A)** the immunoglobulins of milk, which are degraded so the young can absorb other nutrients
- B)** casein, the main protein of milk, which coagulates in the presence of calcium ions
- C)** lactose, the sugar of milk, which is split into glucose and galactose
- D)** milk fat, which is dispersed into fine droplets
- E)** calcium phosphate dissolved in the milk, which precipitates out

Answer: B

Rennin is a protease: it clots casein, the main protein of milk, into a solid curd stabilised by calcium ions, keeping the milk in the stomach longer. The lactose claim confuses rennin with lactase, and the fat claim confuses it with lipase or bile; milk immunoglobulins and dissolved calcium phosphate are not its targets.

96. Which of the following statements about human digestion is false?

- A)** Pancreatic juice supplies proteases, amylase, lipase and nucleases, which act in the small intestine.
- B)** The gallbladder produces bile in response to the hormone cholecystokinin and releases it into the duodenum.
- C)** Pepsin is secreted as the inactive precursor pepsinogen and is activated by the acid environment of the stomach.
- D)** Bile contains no digestive enzymes, yet efficient fat digestion depends on it.
- E)** Salivary amylase starts starch digestion in the mouth, and its action ceases once the food meets the acid stomach.

Answer: B

Bile is produced by the liver; the gallbladder only stores and concentrates it and contracts in response to cholecystokinin, so the statement crediting the gallbladder with producing bile is the false one. Pepsinogen activation by acid, the enzyme content of pancreatic juice, the enzyme-free emulsifying role of bile and the acid-inactivation of salivary amylase are all correctly described.

97. A treatment destroys most of a patient's thyroid gland, including both the follicular cells and the parafollicular (C) cells, while the four parathyroid glands remain intact. Which outcome is expected?

- A)** Blood calcium falls to dangerous levels, because calcitonin was the principal hormone that raises blood calcium.
- B)** Basal metabolic rate rises, because the thyroid hormones are the hormones that slow metabolism down.
- C)** Basal metabolic rate is unaffected, because the thyroid hormones act only on bone remodelling.
- D)** Blood calcium rises uncontrollably, because the loss of calcitonin leaves parathyroid hormone completely unopposed.
- E)** Basal metabolic rate falls, and blood calcium stays close to normal, because parathyroid hormone from the intact parathyroids is the principal regulator of blood calcium.

Answer: E

Step 1: the follicular cells secrete T3 and T4, which set the basal metabolic rate, so destroying them lowers it. Step 2: calcitonin from the C cells only fine-tunes blood calcium downward, while parathyroid hormone from the intact parathyroids is the principal regulator, so blood calcium stays close to normal. Making calcitonin the main calcium-raising hormone reverses its action, and thyroid hormones act on metabolism throughout the body, not only on bone remodelling.

98. Blood tests show that a person's plasma calcium concentration is abnormally high. Which endocrine response is expected?

- A)** Insulin secretion falls, because insulin normally removes calcium from the blood along with glucose.
- B)** Parathyroid hormone secretion rises, stimulating osteoclasts to release further calcium from bone.
- C)** Both parathyroid hormone and calcitonin secretion fall, because both hormones raise blood calcium.
- D)** Parathyroid hormone secretion falls and calcitonin secretion rises, favouring the deposition of calcium in bone.
- E)** Calcitonin secretion falls, which allows calcium to accumulate still further in the blood.

Answer: D

Step 1: abnormally high plasma calcium suppresses parathyroid hormone secretion. Step 2: it stimulates calcitonin release, which inhibits osteoclasts and favours deposition of calcium in bone, restoring the level by negative feedback. Raising parathyroid hormone is the response to low calcium, calcitonin lowers rather than raises blood calcium, and insulin plays no role in calcium homeostasis.

99. In untreated type 1 diabetes, insulin secretion is nearly absent and blood glucose rises far above the normal renal threshold. Why does glucose appear in the urine?

- A)** The filtered glucose load exceeds the transport capacity of the reabsorption mechanisms in the proximal convoluted tubule, so the surplus escapes reabsorption and is excreted.
- B)** Insulin normally seals the glomerular filter against glucose; without it, glucose filters freely and none of it can be reabsorbed.
- C)** The sodium-potassium pumps of the proximal tubule fail, so every reabsorptive process in the nephron stops.
- D)** High blood glucose triggers the active secretion of glucose from the blood into the distal convoluted tubule.
- E)** The collecting duct destroys glucose molecules, so any glucose reaching it can no longer be recovered.

Answer: A

Step 1: glucose is filtered freely at the glomerulus and normally reabsorbed completely in the proximal convoluted tubule by specific sodium-linked co-transporters. Step 2: those transporters have a finite maximum rate. Step 3: in untreated type 1 diabetes the filtered glucose load exceeds that maximum, so the surplus escapes reabsorption and is excreted. Neither the filtration nor this reabsorption depends on insulin, and the nephron neither actively secretes glucose nor destroys it.

100. During a long fast, glucagon helps to keep the blood glucose level stable. Which statement about this hormone is correct?

- A)** It is released by the alpha cells of the pancreas when blood glucose rises after a meal.
- B)** It converts muscle glycogen directly into blood glucose before the liver is involved at all.
- C)** It triggers glycogen breakdown mainly in the liver, while muscle keeps its stores because muscle cells lack glucagon receptors and use their glycogen for their own contraction.
- D)** It triggers glycogen breakdown equally in liver and muscle, so both tissues feed glucose into the blood.
- E)** It promotes the uptake of glucose by muscle cells, lowering the blood glucose level during the fast.

Answer: C

Step 1: glucagon is released by the pancreatic alpha cells when blood glucose falls, as during a fast. Step 2: it acts mainly on the liver, driving glycogenolysis and gluconeogenesis to feed glucose into the blood. Step 3: muscle cells lack glucagon receptors, so muscle glycogen serves only local contraction and never feeds the blood. Release when glucose rises reverses the actual trigger, and glucagon promotes glucose output by the liver, not uptake by muscle.

101. A drug blocks the active pumping of sodium ions out of the ascending limb of the loop of Henle. Which outcome follows?

- A)** The urine becomes maximally concentrated, because sodium accumulates unchecked in the medulla.
- B)** The salt gradient in the medulla weakens, so the collecting duct reabsorbs less water and a large volume of dilute urine is produced, even when ADH is present.
- C)** Glucose escapes into the urine, because the loop of Henle is normally where glucose is reabsorbed.
- D)** ADH secretion rises and concentrates the urine beyond the normal maximum, fully compensating for the block.
- E)** Filtration at the glomerulus stops, because the loop of Henle generates the filtration pressure.

Answer: B

Step 1: active sodium transport out of the ascending limb of the loop of Henle builds the high osmolarity of the medullary interstitium. Step 2: ADH-driven water reabsorption in the collecting duct works only down that gradient. Step 3: with the pump blocked the gradient collapses, so even with ADH present the duct reabsorbs little water and a large volume of dilute urine is produced. ADH cannot act on a gradient that no longer exists, glucose is reabsorbed in the proximal tubule rather than the loop, and glomerular filtration pressure does not depend on the loop of Henle.

102. A drug enters the fluid of the kidney tubule by two routes: filtration at the glomerulus, and active transport from the peritubular blood into the tubule. The second route of entry is called:

- A)** tubular secretion
- B)** reabsorption
- C)** countercurrent multiplication
- D)** dialysis
- E)** pressure filtration

Answer: A

Active transfer of a substance from the peritubular blood into the tubular fluid is tubular secretion, the reverse of reabsorption, which moves substances from the tubule back into the blood. Filtration is the pressure-driven, non-selective process at the glomerulus and cannot describe an active second route, and countercurrent multiplication is the loop-of-Henle mechanism that builds the medullary gradient.

103. A healthy person quickly drinks one litre of pure water and plasma osmolarity falls slightly. Which sequence of events follows?

- A)** Glomerular filtration ceases until plasma osmolarity returns to normal, so no urine is produced.
- B)** ADH release is suppressed, the collecting duct becomes more permeable to water, and urine output falls.
- C)** Aldosterone release stops while ADH stays high, so sodium alone is excreted.
- D)** ADH release is suppressed, the collecting duct becomes less permeable to water, and a large volume of dilute urine is produced.
- E)** ADH release is stimulated, the collecting duct becomes less permeable to water, and a small volume of concentrated urine is produced.

Answer: D

Step 1: the absorbed water slightly dilutes the plasma, and hypothalamic osmoreceptors detect the fall in osmolarity. Step 2: ADH release from the posterior pituitary is suppressed. Step 3: the collecting duct becomes less permeable to water, so the dilute tubular fluid passes through largely unchanged and a large volume of pale urine is produced. Pairing suppressed ADH with increased water permeability reverses the hormone's action, which inserts aquaporins into the duct.

104. Which statement about the cells of bone tissue is correct?

- A)** Chondrocytes resorb calcified matrix in order to release stored calcium.
- B)** Osteocytes are the main cells that form new bone matrix during growth.
- C)** Osteoclasts end up trapped in lacunae as mature cells that maintain the matrix.
- D)** Osteoblasts dissolve bone matrix and release calcium, while osteoclasts build new matrix.
- E)** Osteoclasts resorb bone matrix, releasing calcium and phosphate into the blood, while osteoblasts deposit new matrix.

Answer: E

Osteoclasts resorb bone matrix, releasing calcium and phosphate into the blood, while osteoblasts deposit new matrix; the balance between them, steered by parathyroid hormone and calcitonin, sets blood calcium. The description swapping the two roles is the standard confusion, the matrix-forming cells during growth are osteoblasts rather than osteocytes, cells trapped in lacunae maintaining the matrix are osteocytes rather than osteoclasts, and chondrocytes are cartilage cells that do not resorb calcified matrix.

105. A toxin blocks the voltage-gated potassium channels of a neuron but leaves all other channels and the sodium-potassium pump intact. Which outcome follows when the axon is stimulated above threshold?

- A)** The action potential runs normally, because the rising phase depends on chloride and the falling phase on sodium.
- B)** The membrane repolarises faster than normal, because the sodium channels stay open for longer.
- C)** Depolarisation still occurs, but repolarisation is delayed because K^+ efflux is blocked, so the membrane stays depolarised far longer and cannot carry rapid repeated impulses.
- D)** No depolarisation occurs, because the rising phase of the action potential is driven by potassium leaving the cell.
- E)** The resting potential is lost at once, because the resting membrane potential is maintained by voltage-gated potassium channels.

Answer: C

Step 1: the rising phase of the action potential is driven by voltage-gated sodium channels, so depolarisation still occurs when the potassium channels are blocked. Step 2: repolarisation depends on potassium efflux through voltage-gated potassium channels, and with that efflux blocked the membrane stays depolarised far longer and cannot carry rapid repeated impulses. The resting potential is maintained by non-gated potassium leak channels together with the sodium-potassium pump, so it is not lost at once, and the rising phase has nothing to do with chloride or with potassium leaving.

106. A poison blocks the voltage-gated calcium channels in the presynaptic terminal of a chemical synapse, while impulse conduction along the axon itself is normal. Which outcome follows?

- A)** The postsynaptic cell depolarises normally, because transmission at chemical synapses is electrical rather than chemical.
- B)** The impulse still arrives at the terminal, but vesicle fusion with the membrane is not triggered, so the postsynaptic cell receives no chemical signal.
- C)** The action potential fails to travel along the presynaptic axon, because calcium carries the impulse current there.
- D)** Reuptake of neurotransmitter stops, so the synapse becomes overstimulated by transmitter left in the cleft.
- E)** Neurotransmitter is released continuously, even without impulses, and permanently stimulates the postsynaptic cell.

Answer: B

Step 1: the action potential still travels along the axon and arrives at the terminal, since conduction uses sodium channels. Step 2: calcium entry through voltage-gated calcium channels is the coupling step that triggers fusion of transmitter vesicles with the presynaptic membrane. Step 3: blocked, no transmitter is released and the postsynaptic cell receives no chemical signal. This is not continuous release, which would require calcium to leak in uncontrolled, and reuptake is a separate mechanism from vesicle fusion.

107. A disease progressively destroys the myelin sheath of peripheral motor axons. Which chain of consequences follows for a signal travelling toward the muscle?

- A)** The resting potential reverses along the whole axon at once, producing continuous contraction.
- B)** Neurotransmitter release increases automatically to compensate, keeping muscle activation normal.
- C)** The impulse travels faster, because bare membrane conducts electrical current more readily than insulated membrane.
- D)** The impulse slows and then fails to reach the terminal, because the demyelinated stretches of membrane lack the dense voltage-gated sodium channels needed to regenerate the signal, so muscle activation weakens.
- E)** The impulse continues normally, because saltatory conduction is an optional shortcut rather than a necessity.

Answer: D

Step 1: myelin forces the impulse to jump between the nodes of Ranvier, where the voltage-gated sodium channels are concentrated and regenerate the signal (saltatory conduction). Step 2: demyelinated stretches lack those dense channel clusters, so the local current decays below threshold across them. Step 3: conduction slows and then fails, so muscle activation weakens. Insulation speeds conduction rather than impeding it, saltatory conduction is essential rather than optional, and nothing automatically compensates by releasing more transmitter.

108. A patient is given a drug that mimics acetylcholine at the receptors of the parasympathetic division of the autonomic nervous system. Which set of effects is expected?

- A)** Heart rate falls, pupils constrict, airways widen and digestive activity is inhibited.
- B)** Heart rate rises, pupils constrict, airways narrow and digestive activity is inhibited.
- C)** Heart rate rises, pupils dilate, airways widen and digestive activity is inhibited.
- D)** Heart rate falls, pupils dilate, airways widen and digestive activity is stimulated.
- E)** Heart rate falls, pupils constrict, airways narrow and digestive activity is stimulated.

Answer: E

The parasympathetic division is the rest-and-digest state: it slows the heart, constricts the pupils, narrows the airways and stimulates digestive motility and secretion, so a drug mimicking acetylcholine at parasympathetic receptors produces exactly that set of effects. Raised heart rate, dilated pupils and widened airways are the sympathetic pattern, and each remaining combination mixes one effect of one division with effects of the other.

109. A person walks from bright sunshine into a dim cinema, sees almost nothing at first and regains useful vision over several minutes. Which explanation is correct?

- A)** The pupil reaches its maximum diameter at once, so vision should be sharp immediately.
- B)** Bright light destroyed the cones, and new ones must be produced by cell division before vision returns.
- C)** In bright light the rod pigment rhodopsin was largely bleached and the cones did most of the work; in darkness rhodopsin is resynthesised over minutes and the rods gradually take over.
- D)** Bright light bleached the pigment of the photoreceptors packed in the fovea, where dim vision is normally sharpest.
- E)** In dim light the lens changes shape so that more light is admitted directly into the eye.

Answer: C

Step 1: in bright light the rod pigment rhodopsin is largely bleached and the cones do most of the work. Step 2: in darkness rhodopsin must be resynthesised, a process that takes several minutes, matching the timescale of dark adaptation. Step 3: as rhodopsin returns, the rods take over and useful vision is regained. The fovea contains only cones, so its bleached pigment cannot be what is restored, bright light does not destroy the photoreceptors, and the lens focuses light rather than controlling how much enters.

110. Which statement about the organisation of the human nervous system is correct?

- A)** The central nervous system consists of brain and spinal cord, while the peripheral nervous system divides into the somatic and autonomic systems, the latter carrying sympathetic and parasympathetic branches.
- B)** The autonomic nervous system lies entirely within the brain and spinal cord.
- C)** The sympathetic and parasympathetic divisions are subdivisions of the somatic nervous system.
- D)** The spinal cord belongs to the peripheral nervous system, because peripheral nerves run through it.
- E)** The autonomic nervous system controls the contraction of skeletal muscles during voluntary movements.

Answer: A

The central nervous system is brain plus spinal cord; the peripheral nervous system divides into the somatic system (voluntary, skeletal muscle) and the autonomic system, which carries the sympathetic and parasympathetic branches, and that is the correct organisation. The sympathetic and parasympathetic divisions are not subdivisions of the somatic system, the autonomic system is not confined to the brain and spinal cord, the spinal cord belongs to the central system, and voluntary skeletal-muscle control is somatic, not autonomic.

111. A lumbar puncture is performed to sample the fluid that cushions the brain and spinal cord. The needle passes through the dura mater and then through one further meningeal layer before reaching the fluid. Which layer is pierced last, and which fluid is sampled?

- A)** the arachnoid mater, and the sample is blood from the subdural space
- B)** the dura mater, and the sample is lymph from the epidural space
- C)** the pia mater, and the sample is cerebrospinal fluid from the subdural space
- D)** the pia mater, and the sample is tissue fluid from the subdural space
- E)** the arachnoid mater, and the sample is cerebrospinal fluid from the subarachnoid space

Answer: E

Step 1: the meninges from outside inwards are dura mater, arachnoid mater and pia mater. Step 2: cerebrospinal fluid fills the subarachnoid space between arachnoid and pia. Step 3: after passing the dura, the last layer pierced is the arachnoid mater, and the sample drawn is cerebrospinal fluid. The subdural space normally contains essentially no fluid, and it is neither tissue fluid, lymph nor blood.

112. In an animal experiment both vagus nerves are cut, and the animal is observed after recovery from the surgery. Which set of consequences is expected?

- A)** Heart rate falls, because the vagus carries the main sympathetic acceleration of the heart.
- B)** Heart rate rises above normal, because the tonic parasympathetic slowing exerted by the vagus is lost, while digestive motility and secretion decrease.
- C)** Heart rate is unchanged, because the heartbeat is driven entirely by the sinoatrial node itself.
- D)** Digestive activity accelerates, because the vagus normally inhibits the gut.
- E)** The pupils dilate permanently, because the vagus supplies the muscles of the iris.

Answer: B

Step 1: the vagus carries parasympathetic fibres that impose a tonic brake on the sinoatrial node and stimulate digestive activity. Step 2: cutting both vagi releases the brake, so heart rate rises above normal. Step 3: the same cut withdraws the stimulus to the gut, so digestive motility and secretion decrease. The heart is not driven by the node without autonomic modulation, the vagus excites rather than inhibits the gut, and the iris muscles are not supplied by the vagus.

Section H - Bioenergetics and muscle (Q113-130)

113. Phosphofructokinase-1 (PFK-1) catalyses the committed step of glycolysis. Which of the following statements about its control is false?

- A)** ATP binds to an allosteric regulatory site on PFK-1 and lowers the enzyme's activity.
- B)** A high AMP:ATP ratio in the cytosol favours flux through the PFK-1 step.
- C)** AMP binds to PFK-1 and relieves the inhibition caused by high ATP.
- D)** ATP acts only as an inhibitor of PFK-1 and plays no role whatsoever as its substrate.
- E)** Citrate, an intermediate of the citric acid cycle, allosterically inhibits PFK-1.

Answer: D

PFK-1 uses ATP to phosphorylate fructose-6-phosphate, so ATP is a substrate at the active site, and separately binds to an allosteric regulatory site as an inhibitor at high concentration. The claim denying ATP any substrate role is therefore the false statement. AMP relieves the ATP inhibition, citrate signals abundant biosynthetic precursors and inhibits the enzyme, and a high AMP:ATP ratio does favour flux through this step.

114. During a sprint, the ATP concentration inside a fast muscle fibre falls and its AMP concentration rises. What is the most likely immediate effect on glycolysis in that fibre?

- A)** Glycolysis accelerates, because AMP allosterically activates PFK-1, the enzyme catalysing the committed step.
- B)** Glycolysis slows, because AMP competes with glucose for hexokinase at the entry to the pathway.
- C)** Glycolysis accelerates, because AMP directly phosphorylates ADP to make extra ATP inside the enzyme.
- D)** Glycolysis stops, because PFK-1 requires ATP as an allosteric activator.
- E)** Glycolysis continues at the same rate, because AMP has no effect on glycolytic enzymes.

Answer: A

Step 1: falling ATP and rising AMP signal a low energy charge in the sprinting fibre. Step 2: AMP binds an allosteric site on PFK-1, the enzyme of the committed step, and activates it by relieving ATP inhibition. Step 3: glycolysis accelerates to restore the ATP supply. PFK-1 does not require ATP as an allosteric activator, since high ATP inhibits it, AMP does not compete with glucose at hexokinase, and the activation is allosteric binding, not chemical phosphorylation by AMP.

115. In the aerobic respiration of one glucose molecule, what do the two link reactions (oxidative decarboxylation of pyruvate) together yield, and where do they occur?

- A)** 1 acetyl-CoA, 1 CO₂ and 1 NADH, inside the mitochondrial matrix
- B)** 2 acetyl-CoA, 4 CO₂ and 2 NADH, inside the mitochondrial matrix
- C)** 2 acetyl-CoA, 2 CO₂ and 2 NADH, inside the mitochondrial matrix
- D)** 2 acetyl-CoA, 2 CO₂ and 2 FADH₂, on the inner mitochondrial membrane
- E)** 2 acetyl-CoA, 2 CO₂ and 2 NADH, in the cytosol

Answer: C

Given: aerobic respiration of one glucose; the question covers the two link reactions combined.

Principle: glycolysis splits glucose into 2 pyruvates, and each pyruvate is oxidatively decarboxylated once by the pyruvate dehydrogenase complex in the mitochondrial matrix.

Step 1: per pyruvate: 1 carbon leaves as CO₂, 1 NAD⁺ is reduced to NADH, and the remaining 2-carbon unit joins coenzyme A as acetyl-CoA.

Step 2: multiply by 2 pyruvates per glucose: 2 acetyl-CoA, 2 CO₂, 2 NADH.

Step 3: the complex sits in the mitochondrial matrix, so the products are delivered directly to the citric acid cycle.

Result: 2 acetyl-CoA, 2 CO₂ and 2 NADH, inside the mitochondrial matrix.

Classic slip: counting 4 CO₂ imports the two decarboxylations of the citric acid cycle itself into the link reaction.

116. A toxin irreversibly inhibits the pyruvate dehydrogenase complex in cells that are supplied only with glucose. Which outcome follows?

- A)** Pyruvate accumulates, acetyl-CoA production falls and the ATP yield from glucose oxidation drops sharply.
- B)** Acetyl-CoA accumulates and activates the citric acid cycle, raising ATP output.
- C)** The electron transport chain accelerates, because NADH from glycolysis alone floods into the mitochondria.
- D)** Glycolysis stops, because pyruvate dehydrogenase catalyses an essential step of glycolysis itself.
- E)** Pyruvate accumulates but the citric acid cycle speeds up to compensate, keeping ATP output normal.

Answer: A

Step 1: glycolysis still runs, so pyruvate is produced but can no longer be converted to acetyl-CoA and accumulates. Step 2: the citric acid cycle is starved of acetyl-CoA, so NADH and FADH₂ generation falls. Step 3: the electron transport chain runs short of electrons and the ATP yield from glucose oxidation drops sharply. The cycle cannot compensate for losing its input, the blocked enzyme is not part of glycolysis itself, and the chain slows rather than accelerates.

117. What are the products of one complete turn of the citric acid cycle?

- A)** 2 NADH, 2 FADH₂, 1 GTP and 2 CO₂
- B)** 3 NADH, 1 FADH₂, 1 GTP and 2 CO₂
- C)** 3 NADH, 1 FADH₂, 1 ATP by oxidative phosphorylation and 2 CO₂
- D)** 3 NADH, 1 FADH₂, 1 GTP and 1 CO₂
- E)** 4 NADH, 1 FADH₂, 1 GTP and 1 CO₂

Answer: B

Given: one complete turn of the citric acid cycle, starting from one acetyl-CoA joined to oxaloacetate.

Principle: count each redox, decarboxylation and phosphorylation step of the cycle separately.

Step 1: three NAD⁺-reducing steps (isocitrate oxidation, alpha-ketoglutarate oxidation, malate oxidation) give 3 NADH.

Step 2: the single FAD-dependent step (succinate to fumarate) gives 1 FADH₂.

Step 3: succinyl-CoA synthetase gives 1 GTP by substrate-level phosphorylation.

Step 4: the two decarboxylation steps (of isocitrate and of alpha-ketoglutarate) release 2 CO₂.

Result: per turn: 3 NADH, 1 FADH₂, 1 GTP and 2 CO₂.

Classic slip: counting 2 FADH₂ double-books the single succinate dehydrogenase step, and the GTP is substrate-level, not a product of oxidative phosphorylation.

118. Acetyl-CoA labelled with ^{14}C on its methyl (CH_3) carbon enters the citric acid cycle. When is the label first released as $^{14}\text{CO}_2$?

- A)** During the first turn of the cycle, as one of the two CO_2 molecules released
- B)** Never, because the methyl carbon is fully conserved for as long as the cycle runs
- C)** During the second turn, after the label has passed into oxaloacetate
- D)** From the third turn onwards: neither carbon of the entering acetyl group is released during the first turn, and the methyl carbon survives even longer than the carbonyl carbon
- E)** Immediately on entry, because citrate synthase decarboxylates acetyl-CoA as it condenses

Answer: D

Step 1: in any turn of the cycle, the two CO_2 released come from carbons that belonged to the oxaloacetate, not to the acetyl group that just entered. Step 2: an incoming acetyl group therefore completes its first full turn intact. Step 3: the label passes into oxaloacetate; the carbonyl carbon of the acetyl group is released as CO_2 mainly in the second turn, while the methyl carbon survives even longer and appears as $^{14}\text{CO}_2$ mainly from the third turn onwards. Citrate synthase condenses acetyl-CoA with oxaloacetate without any decarboxylation, so release cannot happen on entry, and the label is not conserved indefinitely.

119. Which of the following statements about the citric acid cycle is false?

- A)** The cycle runs mainly in the mitochondrial matrix, with succinate dehydrogenase embedded in the inner membrane.
- B)** Acetyl-CoA condenses with oxaloacetate, which is regenerated at the end of every turn.
- C)** Each turn of the cycle consumes two molecules of acetyl-CoA and regenerates one molecule of oxaloacetate.
- D)** The GTP formed in the cycle is produced by substrate-level phosphorylation.
- E)** Succinate dehydrogenase transfers electrons from succinate to FAD, and the resulting FADH_2 passes them on directly within the respiratory chain.

Answer: C

Given: statements about citric acid cycle organisation; one claims that each turn consumes two acetyl-CoA.

Principle: the cycle is a closed loop whose oxaloacetate carrier is regenerated, so acetyl groups must be counted per turn and per glucose separately.

Step 1: one turn condenses exactly 1 acetyl-CoA with 1 oxaloacetate and returns 1 regenerated oxaloacetate.

Step 2: one glucose yields 2 acetyl-CoA, which enter the cycle in 2 separate turns, so 2 is a per-glucose total, not a per-turn consumption.

Result: the two-acetyl-CoA-per-turn claim is false; every turn consumes one acetyl-CoA and regenerates the oxaloacetate.

Classic slip: the true statement that one glucose feeds two acetyl groups into the cycle is misread as a property of a single turn.

120. Which of the following statements about the mitochondrial electron transport chain is false?

- A)** Complex II transfers electrons from succinate, via FADH_2 , to ubiquinone without pumping protons.
- B)** Complex III passes electrons from ubiquinol to cytochrome c and pumps protons.
- C)** Complex I transfers electrons from NADH to ubiquinone and pumps protons to the intermembrane space.
- D)** Complex IV transfers electrons to O_2 , reducing it to water, and pumps protons.
- E)** The final electron acceptor for the electrons carried by NADH is cytochrome c of complex III.

Answer: E

Electrons from NADH end on molecular oxygen, which complex IV reduces to water; cytochrome c is only a mobile shuttle between complexes III and IV, so calling it the final acceptor is the false statement. Complex I does pass electrons from NADH to ubiquinone while pumping protons, complex II feeds succinate-derived electrons to ubiquinone without pumping, which is why FADH_2 bypasses complex I, and complexes III and IV pump as stated.

121. Rotenone specifically blocks electron transfer at complex I of the respiratory chain. A respiring cell is then supplied with succinate as its main oxidisable substrate. Which outcome is expected?

- A)** ATP synthesis stops completely, because every electron entering the chain must first pass through complex I.
- B)** ATP synthesis continues at a reduced rate, because electrons from succinate FADH_2 enter the chain at complex II and drive pumping at complexes III and IV.
- C)** ATP synthesis continues at the normal rate, because complex II pumps protons and takes over the work of complex I.
- D)** The proton gradient rises above its normal value, because succinate oxidation pumps more protons than NADH oxidation.
- E)** Oxygen consumption stops completely, even though the proton gradient persists.

Answer: B

Step 1: succinate hands its electrons to ubiquinone through complex II, which lies downstream of the blocked complex I. Step 2: from ubiquinone the electrons flow through complexes III and IV, which still pump protons, so ATP synthesis continues, though at a reduced rate because the contribution of complex I is lost. Complex II pumps no protons, so the gradient cannot rise above normal, electrons from succinate need not pass through complex I, and oxygen consumption continues while succinate feeds the chain.

122. Cyanide binds the Fe^{3+} of cytochrome a_3 in complex IV and halts electron transfer to oxygen. In a mitochondrion poisoned in this way, what happens?

- A)** Complexes I, II and III become fully oxidised while NADH accumulates in the matrix.
- B)** The proton gradient is unaffected, because ATP synthase can still run in reverse.
- C)** Proton pumping continues at complexes I and III even though complex IV is blocked.
- D)** Oxygen consumption continues at a lower rate while NAD^+ is steadily regenerated.
- E)** Complexes I, II and III become fully reduced, proton pumping stops and ATP synthesis falls to zero.

Answer: E

Step 1: with cytochrome a_3 blocked, electrons can no longer pass to oxygen, so the whole chain backs up behind the block. Step 2: complexes I, II and III become fully reduced, proton pumping stops everywhere, and the proton gradient drains until ATP synthesis falls to zero. The mirror-image description, upstream carriers fully oxidised while NADH accumulates, states exactly what cannot happen, since oxidising NADH is precisely what stops, and ATP synthase running in reverse cannot preserve the gradient.

123. A molecule that makes the inner mitochondrial membrane leaky to protons is added to actively respiring mitochondria. Which outcome is expected?

- A)** Electron transport stops, because the proton gradient is the driving force for electron flow.
- B)** ATP synthesis increases, because protons now return to the matrix faster.
- C)** Electron transport and oxygen consumption speed up, most of the released energy appears as heat, and ATP synthesis falls.
- D)** Electron transport slows down, while ATP synthesis is unaffected.
- E)** ATP synthesis continues at its normal rate, because ATP synthase is not part of the electron transport chain.

Answer: C

Step 1: the proton leak dissipates the gradient that constrains electron flow, so the chain runs faster and oxygen consumption rises. Step 2: the energy stored in the gradient escapes as heat instead of driving ATP synthase. Step 3: ATP synthesis falls even though electron transport speeds up, the principle behind heat production in brown fat. Electron flow is limited, not driven, by the gradient, and protons returning through a leak do not make ATP.

124. Per molecule delivered to the respiratory chain, FADH_2 yields roughly two ATP, whereas NADH yields roughly two to three times as many. What is the main reason for the difference?

- A)** FADH_2 donates its electrons directly to oxygen at complex IV.
- B)** FADH_2 carries only one electron, whereas NADH carries two.
- C)** FADH_2 releases its electrons into the intermembrane space, where they cannot reach the chain.
- D)** FADH_2 enters the chain at complex II, so its electrons bypass complex I and its oxidation pumps fewer protons.
- E)** FADH_2 cannot be reoxidised inside the mitochondrion and is excreted from the cell.

Answer: D

Step 1: NADH delivers its electrons to complex I, which pumps protons before handing them to ubiquinone. Step 2: FADH_2 delivers its electrons to complex II, which passes them to ubiquinone without pumping. Step 3: FADH_2 electrons therefore skip one proton-pumping step, store less energy in the gradient and yield roughly two ATP instead of about three. Both carriers hold two electrons, so an electron-count difference cannot explain the yield gap, and FADH_2 neither donates to oxygen directly nor is excreted.

125. Which of the following statements about the roles of NAD^+ , FAD and ATP in metabolism is correct?

- A)** NAD^+ and FAD accept electrons, together with protons, from oxidised substrates, while ATP transfers usable energy mainly through its terminal phosphate bond.
- B)** ATP accepts electrons from oxidised substrates and passes them on to the electron transport chain.
- C)** NAD^+ and FAD store the energy they harvest as a proton gradient inside their own rings.
- D)** ATP is reduced to ADP when it transfers its terminal phosphate group to a substrate.
- E)** FAD accepts two electrons and two protons during glycolysis, while NAD^+ accepts single electrons only.

Answer: A

NAD^+ and FAD accept electrons, together with protons, from oxidised substrates, becoming NADH and FADH_2 that feed the respiratory chain, while ATP transfers usable energy mainly through its terminal phosphate bond: two distinct carrier roles, which is the correct statement. ATP does not transport electrons, phosphate transfer is not a reduction of ATP, NAD^+ accepts a hydride equivalent of two electrons rather than single electrons, and the harvested energy is not stored inside the carriers' rings.

126. Which of the following statements about alcoholic fermentation in yeast is correct?

- A)** The NADH that reduces acetaldehyde is supplied by the electron transport chain running in reverse.
- B)** Pyruvate is reduced directly to ethanol, so no CO₂ is produced.
- C)** Alcoholic fermentation generates extra ATP beyond glycolysis by substrate-level phosphorylation.
- D)** Yeast regenerates NAD⁺ during fermentation using lactate dehydrogenase.
- E)** Pyruvate is first decarboxylated to acetaldehyde with release of CO₂, then reduced to ethanol by NADH, regenerating NAD⁺ with no further ATP synthesis.

Answer: E

Step 1: pyruvate decarboxylase converts pyruvate to acetaldehyde with release of CO₂, the gas of fermentation. Step 2: alcohol dehydrogenase reduces acetaldehyde to ethanol using the NADH made by glycolysis, regenerating NAD⁺ so glycolysis can continue. Step 3: neither fermentation step makes ATP; the pathway's only ATP comes from glycolysis itself. Pyruvate is not reduced directly to ethanol without CO₂ release, lactate dehydrogenase is the muscle route, and the NADH used comes from cytosolic glycolysis, not from a reversed electron transport chain.

127. A muscle fibre metabolises glucose entirely by glycolysis followed by lactic acid fermentation. What is the net ATP yield per glucose, and why does fermentation remain indispensable?

- A)** 0 ATP: glycolysis cannot even start without oxygen.
- B)** 2 ATP: fermentation regenerates the NAD⁺ that glycolysis needs at the step oxidising glyceraldehyde-3-phosphate.
- C)** 4 ATP: the fermentation steps themselves produce two extra ATP per pyruvate.
- D)** 6 ATP: fermentation converts the NADH of glycolysis into ATP inside the mitochondria.
- E)** 8 ATP: the conversion of pyruvate to lactate feeds electrons into the electron transport chain.

Answer: B

Given: one glucose metabolised by glycolysis followed by lactic acid fermentation, no oxygen available.
Principle: net ATP = ATP invested - ATP recovered, and glycolysis can continue only if the NADH it produces is reoxidised to NAD⁺.

Step 1: the investment phase spends 2 ATP per glucose.

Step 2: the payoff phase recovers 4 ATP per glucose (2 per pyruvate, 2 pyruvates).

Step 3: net = 4 - 2 = 2 ATP per glucose

Step 4: the NADH from the glyceraldehyde-3-phosphate step reduces pyruvate to lactate, regenerating the NAD⁺ that step requires, so glycolysis keeps running without oxygen.

Result: net yield 2 ATP per glucose, with fermentation indispensable as the NAD⁺-regenerating step.

Classic slip: the fermentation steps themselves produce no ATP, so any total above 2 double-counts glycolysis or invents a mitochondrial step.

128. During skeletal muscle contraction, what is the role of ATP at the moment a myosin head detaches from actin?

- A)** ATP binding to the myosin head weakens its grip on actin and causes detachment; hydrolysis of that ATP then re-cocks the head for the next stroke.
- B)** Calcium released by ATP detaches myosin from actin.
- C)** ATP phosphorylates actin so that myosin can bind it again at a fresh site.
- D)** Detachment requires no ATP, because the release of ADP alone breaks the cross-bridge.
- E)** ATP is consumed only at the end of the power stroke, to pull the filaments past each other.

Answer: A

Step 1: after the power stroke the myosin head remains locked onto actin. Step 2: binding of a fresh ATP to the head weakens its grip and detaches the cross-bridge. Step 3: hydrolysis of that ATP re-cocks the head into its high-energy position for the next stroke. This is why muscles stiffen in rigor mortis when ATP is gone, and calcium's role is regulatory, exposing the myosin-binding sites through troponin and tropomyosin rather than detaching heads.

129. Dystrophin is absent or non-functional in Duchenne muscular dystrophy. What is its normal role in a skeletal muscle fibre?

- A)** It binds calcium to troponin, exposing the myosin-binding sites on actin.
- B)** It stores oxygen inside the fibre, as myoglobin does.
- C)** It anchors the actin cytoskeleton to membrane glycoproteins, stabilising the sarcolemma during contraction.
- D)** It forms the thick filaments of the sarcomere together with myosin.
- E)** It hydrolyses ATP to supply the power stroke of myosin.

Answer: C

Dystrophin links the actin cytoskeleton beneath the sarcolemma to membrane glycoproteins, stabilising the fibre membrane against the mechanical stress of repeated contraction. Without it the sarcolemma tears, calcium leaks in and the fibres degenerate, which is the basis of Duchenne muscular dystrophy. ATP hydrolysis is myosin's job, oxygen storage is myoglobin's, calcium binding to troponin is the regulatory step, and the thick filaments are made of myosin, so none of those options describes a structural membrane linker.

130. A muscle biopsy from an athlete shows fibres rich in mitochondria and myoglobin, with few glycogen granules and strong resistance to fatigue. Which fibre type is this, and in which sport would such fibres predominate?

- A)** Type II fast glycolytic fibres, as in a marathon runner
- B)** Cardiac muscle fibres, as in a long-distance swimmer
- C)** Type I slow oxidative fibres, as in a sprinter
- D)** Type I slow oxidative fibres, as in a marathon runner
- E)** Type II fast glycolytic fibres, as in a sprinter

Answer: D

Dense mitochondria and abundant myoglobin with few glycogen granules and strong fatigue resistance are the signature of type I slow oxidative fibres, built for continuous aerobic work, and such fibres predominate in endurance athletes such as marathon runners. The sprinter is the trap: fast glycolytic fibres rely on anaerobic glycolysis, are relatively pale and fatigue rapidly, and cardiac muscle is a distinct tissue rather than a skeletal fibre type.